

Capability Scaling-Down Laws of LLM Compression in Agent Systems

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Abstract

Specialized workloads need specific capabilities, and the models that serve them are usually dense sources scaled down by pruning, quantization, or distillation; choosing a configuration today means measuring each one. We develop capability-conditioned empirical models of the compression response: for math, code, and multi-hop QA we predict the conditional loss after compression from the initial model state (parameter count, pretraining tokens, and the dense capability losses of the source or initial student) and the configuration of the method. Relations are built on a controlled panel of Pythia checkpoints whose size and pretraining stage vary independently and validated with predictions frozen before new configurations and states are measured. A five-parameter pruning form predicts math and code at unseen densities inside the fitted range on three new checkpoints (0.24 nats mean absolute error against 0.45 for a source-free strength curve) and fails beyond the range and for QA; a frozen response surface in bit-width and group size predicts unseen quantization configurations on the development states; under a fixed distillation protocol a response model in training budget, reuse, and a student descriptor predicts unseen data pools on the development students, while joint transfer to a larger student and a new pool remains unreliable. We connect the relations to an explicit capability- and budget-specific selection problem, validated retrospectively on held-out states, and release coefficients, tested ranges, registered predictions, and measurements.

1 Introduction

Deployed language-model agents are specialists: a coding agent lives on code synthesis, a retrieval assistant on multi-hop question answering, and both pay for every token of inference. The model that serves such a workload is usually a dense source scaled down by pruning, quantization, or distillation, and choosing the route and its strength today means running each candidate through capability benchmarks. A single compression ratio cannot replace that search, because compression does not shrink capabilities uniformly: across twelve public models from five series, the same pruning density leaves QA the least-damaged capability in some series and the most-damaged in others, and 3-bit quantization collapses twelve of fourteen models (Appendix F).

We therefore ask how the capability-conditioned loss after compression can be predicted from the initial model state and the configuration of the method. The endpoint is the conditional loss of each capability on a fixed probe set, written relative to a reference measured beforehand, the dense source for pruning and quantization and the initial student for distillation (Eq. equation 2). The admitted inputs are the parameter count, the pretraining token count where disclosed, the dense capability losses of the reference, and the intervention setting; no measurement of the compressed target enters a prediction. The relations are developed on a controlled panel of Pythia checkpoints whose size and pretraining stage vary independently, by comparing low-dimensional forms that use the same inputs, and tested by freezing coefficients and predictions before new configurations and states are measured. This separates whether the initial-state inputs help, whether a form adds anything over a same-input alternative, and how far the relation transfers.

The outcome is a set of method-specific relations with measured ranges. For pruning, a five-parameter power form with a source-conditioned amplitude predicts math and code at unseen densities inside the fitted range, ties a per-density regression, and fails beyond the range and for QA. For quantization, bit-width alone supports only a per-bit source regression, and with the group size as a second axis a frozen low-order response surface predicts unseen bit-widths and group sizes on the development states. For distillation, under a fixed protocol a response model in training budget, reuse count, and a student descriptor improves prediction for unseen data pools on the two development students, while joint transfer to a larger student and a new pool remains unreliable. Our contributions are (1) the prediction problem, its admitted inputs, and a development-then-validation design that separates input value, form value, and transfer range; (2) the per-method relations, their coefficients, and their independent tests against same-input and source-free baselines, with the capabilities and settings they cover and where they stop; and (3) a connection of these relations to an explicit capability- and budget-specific selection problem, validated retrospectively on held-out states, with measurements, registered predictions, and code released.

2 Experimental Setup

2.1 Capability-Conditioned Loss

We measure three capabilities, mathematical reasoning, code generation, and multi-hop question answering, each through a fixed probe set $\mathcal{P}_c$ of prompt and reference pairs drawn once from held-out splits of MATH-500, MBPP, and 2WikiMultihopQA. The observable is the token-weighted conditional cross-entropy of the reference completion,

$$L_c(w) = \frac{1}{T_c} \sum_{(x,y) \in \mathcal{P}_c} \sum_{t=1}^{|y|} -\log p_w(y_t \mid x, y_{<t}), \quad T_c = \sum_{(x,y) \in \mathcal{P}_c} |y|, \quad (1)$$

with prompt tokens masked, sequences truncated to 1024 tokens, the model’s own tokenizer, and the same 64 measurement probes per capability for every state compared within a model. Responses are within-model differences on identical token sequences in native nats per token: for pruning and quantization the damage $\Delta L_c = L_c(\text{compressed}) - L_c(\text{dense})$, for distillation the signed transfer $\delta_c = L_c(S_{\text{KD}}) - L_c(S_0)$ of a student relative to its own initial state. The observable is the likelihood of a fixed reference on a fixed distribution: a QA loss can fall while exact-match accuracy does not rise, and a gain on one QA distribution need not appear on another. Absolute levels of different tokenizers are never placed on one scale, and capability specificity is a tested property (Appendix E).

2.2 Models and Compression Settings

A heterogeneous panel of twelve finished public models from five series (Qwen3, Gemma-3, Gemma-4, Muse, OLMo-3; 0.6B to 32B) establishes the phenomena and the limits of cross-family transfer (Appendix F). A controlled panel of Pythia checkpoints (160M, 410M, 1.4B at steps 16k, 64k, 143k) varies the initial state along parameter count and pretraining stage with architecture and data fixed and is the development set of every relation. New states and configurations that entered no fit (Pythia 160M to 6.9B at further steps, new densities, bit-widths, and group sizes, and Gemma-3 270M, 1B, and 4B students at new pools) provide the validation; Appendix A lists them.

The three operations are held fixed by protocol. Pruning is global magnitude pruning of all transformer weight matrices to a retained density d . Quantization is symmetric round-to-nearest of the same matrices, weights only, per output channel at bit-width b or with contiguous input groups of g weights sharing one scale. Distillation is supervised fine-tuning of an existing student on black-box teacher traces with strict LoRA adapters from a pool of U traces per domain, processed for a fixed budget. The inputs admitted to every predictor are the parameter count N_0 of the transformer weight matrices, the pretraining token count D_0 where disclosed, and the dense capability losses $L_{0,c}$ measured once on the uncompressed model (the initial student for distillation). Every prediction is written as

$$\hat{L}_{m,c} = L_{0,c}^{\text{obs}} + \hat{\Delta}_{m,c}(\mathbf{x}, \boldsymbol{\theta}_m), \quad \mathbf{x} = (N_0, D_0, L_{0,c}), \quad (2)$$

with θ_m the setting of method m (density, bit-width and group size, or pool and budget). The function $\hat{\Delta}_{m,c}$ is method-specific; the arms share the endpoint and the evaluation protocol, and each keeps its own functional form.

2.3 Fitting and Evaluation

Each relation is fit on development states and settings only and is judged on unseen settings (a density, bit-width, group size, pool, or budget outside the fitted set) and on unseen states (a new size, stage, or student). On the development panel we use leave-one-stage-out and leave-one-source-out cross-validation; for new states the forms, coefficients, and predictions of every candidate are committed to the repository before the outcome is measured, and a prediction keeps its provenance label whatever the outcome. The dense anchor $L_{0,c}$ is the only measured input. Error is the mean absolute error of $\hat{\Delta}$ in nats per token per capability; every candidate is compared on the same cells with source-free references (zero change, a per-capability constant, a strength-only curve, or the median development curve), with same-input alternatives, and with the no- D_0 ablation, and the strongest baseline per capability is reported and marked as chosen after the fact. Freeze commits, provenance codes, fold rules, and bootstrap units are in Appendix I.

3 Developing Capability-Conditioned Scaling Laws

3.1 Conditioning on the Initial Model State

The first question is whether the initial state is worth reading, and which part of it carries the signal. On the nine-state panel with one pretraining stage held out, a per-setting regression of the pruning response on $(N_0, D_0, L_{0,c})$ lowers the error from 0.413, 0.516, and 0.500 nats (the strongest source-free reference) to 0.253, 0.292, and 0.806 for math, code, and QA; the improvement interval excludes zero for code, includes zero for math, and is negative for QA, where predicting no change is better. Adding the dense anchor $L_{0,c}$ to (N_0, D_0) has intervals above zero for math (+0.55, [0.29, 1.00]) and code (+0.66, [0.38, 1.13]); adding D_0 to $(N_0, L_{0,c})$ does not (Fig. 3A), and a frozen test at a new stage confirmed the ranking. Every relation is therefore conditioned on the three inputs, with the dense anchor as the carrier and D_0 as a weak second-order term, per capability.

3.2 Pruning

Pruning damage grows with the removed fraction $1 - d$ with a source-dependent amplitude, and QA can improve while math and code degrade (Fig. 1, left). The delivered form is

$$\widehat{\Delta L_c}(\mathbf{x}, d) = A_c(\mathbf{x}) \left(\frac{1-d}{0.3} \right)^{\gamma_c}, \quad A_c(\mathbf{x}) = \beta_c \cdot (1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0), \quad (3)$$

five parameters per capability, tildes standardized on the development set (Appendix A, Table 3). Its same-input alternatives are the per-density regression A2 (four coefficients at each development density, an unseen d by linear interpolation), the power form with $\gamma_c = 1$, and a two-term polynomial in $1 - d$; the source-free references are a strength-only curve, the median development curve, and zero change. Fit on the nine states at $d \in \{0.9, 0.8, 0.7, 0.6\}$ and evaluated at the unseen 0.65 and 0.55, Eq. equation 3 cuts the strength-only error by two to three times on every capability, and A2 is better on math and code and within noise on QA (Table 1): reading the source state is what matters, and the power structure adds nothing over the per-density regression. After the frozen tests of §4 both were refit on 17 states, including the unblinded 1B and 6.9B checkpoints, under a pre-committed leave-one-source-out rule: A2 (0.479) and the power form (0.494) tie within 0.02 nats, and the rule prefers the continuous form that vanishes at $d = 1$. Its coefficients are in Appendix B; γ_c is a fitted constant per capability whose value these data do not establish as necessary.

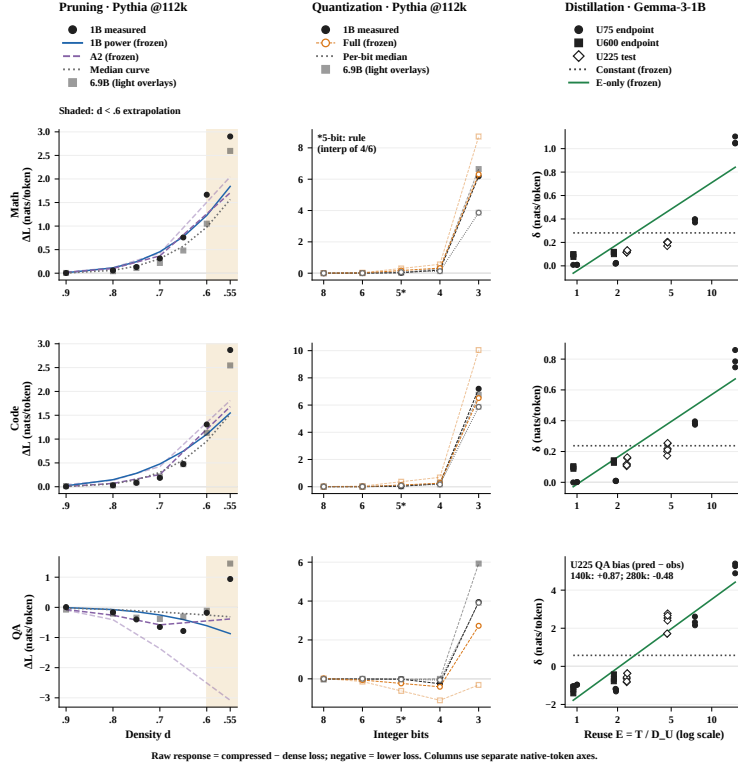


Figure 1: Capability-conditioned compression responses and frozen predictions, one source per arm. Left: pruning of Pythia-1B at step 112k (measured, filled circles) with the frozen power form (Eq. 3), the per-density regression A2, and the source-free median curve; the 6.9B source is overlaid in grey and the extrapolated density 0.55 is shaded. Middle: per-channel quantization of the same sources with the frozen per-bit regression (hollow) and the per-bit median (dotted). Right: distillation of Gemma-3-1B, signed transfer δ_c against the reuse count E ; endpoint pools (fit) and the unseen pool U225 (test) with the frozen constant and reuse-count forms. Lines are fits frozen before the test points were measured; native nats per token.

3.3 Quantization

Bit-width alone offers little room for a continuous relation: int8 and int6 stay within 0.01 nats of dense, int4 costs 0.03 to 0.35 nats on most states, and int3 collapses most models by several nats (Fig. 1, middle). At fixed bit-width we use the source regression of \$3.1 per bit; it predicts a new state at a seen bit-width, is no law over b , and improves on the per-bit median for math and code and not for QA. A grouped quantizer whose scale is shared by g contiguous weights changes the error at a fixed bit-width and is known before the operation, so it adds a genuine configuration axis: on six development states at $b \in \{3, 5\}$ and $g \in \{64, 256\}$, 5-bit cells stay within 0.1 nats and 3-bit damage roughly halves when g drops from 256 to 64. Three relations were fit on those 24 cells with the same inputs: a separable form $A_c(\mathbf{x}) q_{\max}^{-P_c} (g/128)^{q_c}$, the low-order two-dimensional form

$$\widehat{\Delta L}_c(\mathbf{x}, b, g) = \phi(\mathbf{x}) \cdot [1, u, v, uv, u^2], \quad u = \log_2 q_{\max}(b) \text{ (centred)}, \quad v = \log_2 \frac{g}{128}, \quad (4)$$

with one four-vector of state coefficients per term, and a same-input bilinear interpolation. Because b takes two development values, u^2 is constant on the development cells and the design of Eq. equation 4 has rank 16 for its 20 coefficients: the curvature in u at an unseen bit-width is set by the ridge penalty and is not identified by the data (Appendix C). With six states the leave-one-state-out errors are large for all three forms, so the development

set selects no form; the frozen tests of §4.1 do, and they can establish only that a frozen response surface improves prediction at unseen configurations.

3.4 Distillation

The response is the signed transfer δ_c of an existing student to teacher traces, relative to the student’s own dense loss; the teacher never enters it. Its coordinates are the supervised completion tokens T , the pool size D_U , and the reuse count $E = T/D_U$. QA improves under every recipe we ran while math and code respond by a few hundredths to a few tenths of a nat, and with a small pool the QA loss rises again once E grows large (Fig. 1, right). Distillation lowers the conditional loss on the primary QA distribution, and that change does not carry over to TriviaQA on the same checkpoints (Appendix E); we therefore read the QA result as a prediction relation on the measured QA distribution and not as a general QA gain.

Source axis. On the nine Pythia states as students, with the teacher pool and budget fixed, a linear form $\delta_c = \beta_c \cdot (1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0)$ halves the error of a per-capability constant for math (0.014 against 0.031 with one stage held out), ties it for code (0.057 against 0.061), and is worse than a constant for QA (0.263 against 0.159): the student state carries the math response.

Pool, budget, and student. Under the uniform absolute-exposure protocol (two development students, two pools, three pool samples, four fixed checkpoints), eight forms per capability in T , E , and a source term $k \log(N_S/N_{\text{ref}})$ were frozen before any test run. The headline form, the joint form $\hat{\delta}_c = u(a_c + k_c n + b_c u + q_c v)$ with $u = \log(1 + T/T_{\text{ref}})$, $v = \log(D_U/D_{\text{ref}})$, and $n = \log(N_S/N_{\text{ref}})$, is the frozen form with the lowest development error; that selection rule was stated after the tests, so the candidates’ predictions are prospective and the headline choice is retrospective. On the nine test trajectories at an unseen pool its gain over the frozen constant is detected on five of the six student-capability cells of the two development students, and on the held-out 4B student the constant is better at the unseen pool (§4). Two forms that vanish at $T = 0$ and carry a student descriptor $z_c = \tilde{L}_{0,c}$,

$$\hat{\delta}_c = (a_c + \lambda_c z_c) \log(1 + E), \quad \hat{\delta}_c = (a_c + \lambda_c z_c)(1 - e^{-T/T_*}) + (b_c + \mu_c z_c) \log(1 + E), \quad (5)$$

were compared on the development matrix by leave-one-run-out cross-validation and scored on the test matrix retrospectively (Appendix C): the second is the best of all forms on the 270M unseen-pool trajectories, the first fails for QA because it has no budget term, and neither beats the frozen constant for math and code on the 4B student. The delivered relation is the joint form inside the development size range; no joint student-pool-budget law is claimed.

4 Generalization to Unseen Models and Configurations

Once a form is chosen, its coefficients are frozen and it predicts three kinds of unseen object: compression settings outside the fitted set, model states of new sizes or stages, and, across both, the three capabilities. Table 1 lists every registered test with the candidate error and the strongest relevant baseline; Fig. 2 shows where the candidates gain over a source-free reference. Errors are MAE in nats per token.

4.1 Prediction at Unseen Compression Settings

Pruning densities. On the three checkpoints that entered no fit, at densities 0.85, 0.675, and 0.575, the delivered power form has MAE 0.24, 0.24, and 0.68 for math, code, and QA: it beats the strength-only curve (0.45, 0.49, 0.58) and zero change on every checkpoint, has the lowest math error of all predictors, ties the median curve on code, and loses QA to every source-free curve (median 0.22) because it over-predicts the QA improvement, so the median has the lowest aggregate error (0.24 against 0.39). At $d = 0.55$, beyond the range, every source-conditioned form fails on every source (0.6 to 4.5 nats): the transfer range is the interpolation regime.

Test	unseen	origin	math	code	QA
Source axis: seen setting, unseen source state					
prune: fixed d , 3×3 , leave-one-stage-out	stage	RFRA	0.253/0.413 _{med}	0.292/0.516 _{med}	0.806/0.500 _o
quant: fixed b , 3×3 , leave-one-stage-out	stage	RFRA	1.144/1.410 _{med}	0.626/1.553 _{med}	1.989/1.473 _{med}
prune: stage 96k (160M, 1.4B), all d	stage	PFRA	0.090/0.343 _{nD}	0.128/0.179 _{nD}	0.426/0.438 _{nD}
quant: new stage 96k (160M/410M/1.4B), $b \geq 4$	stage	P-RA	0.029/0.084 _{pbm}	0.104/0.094 _{pbm}	0.173/0.134 _{pbm}
distill: δ_c , 3×3 LoRA, leave-one-stage-out	stage	RFRA	0.014/0.026 _{nD}	0.057/0.044 _{nD}	0.263/0.159 _c
Configuration axis: unseen setting, size, stage, or pool					
prune: d 0.65/0.55, dev sources + 410M@96k	d	PRFA	0.371/0.283 _{A2}	0.524/0.279 _{A2}	0.759/0.818 _{A2}
prune: 1B@32k+112k, d 0.9–0.6	size, stage	PFFA	0.215/0.169 _{so}	0.191/0.119 _{A2}	0.261/0.227 _{med}
prune: 1B@32k+112k, $d = 0.55$	size, stage, d	PFFA	1.790/0.939 _{so}	1.488/0.089 _{so}	2.225/0.393 _{so}
prune: 6.9B@32k+112k, d 0.9–0.6	size $\uparrow$, stage	PFFA	0.299/0.120 _{med}	0.160/0.144 _{A2}	0.909/0.120 _{med}
quant: 6.9B@32k+112k, $b \geq 4$	size $\uparrow$, stage, b	PFFA	0.181/0.036 _{pbm}	0.177/0.043 _{pbm}	0.428/0.034 _o
distill: unseen pool U225 (Gemma-3-1B)	pool	PFFA	0.191/0.122 _c	0.117/0.069 _c	0.706/0.710 _{TE}
KD 270M, pool 375	pool	P/F/F/A	0.045/0.060 _c	0.036/0.047 _c	0.606/0.862 _c
KD 1B, pool 375	pool	P/F/F/A	0.050/0.057 _c	0.029/0.069 _c	0.444/1.004 _c
KD 4B, pool 375	student, pool	P/F/F/A	0.089/0.020 _c	0.050/0.038 _c	1.141/1.024 _c

Table 1: Headline prediction tests, all three arms: MAE in nats per token per capability, candidate/strongest baseline (bold = better; c, med, so, 0 = constant, median, strength-only, and zero-change baselines). Origin code: prediction · same-input baseline · simple baseline · scoring, with P/F/A prospective or pre-specified and R retrospective. Complete list and full legend: Table 12 in Appendix D.

Quantization settings. The per-channel predictor’s one unseen setting, 5 bits by a fixed interpolation rule, is worse than a per-bit constant on the 1B source. With group size as a second axis, the frozen surface of Eq. equation 4 has MAE 0.19, 0.21, and 0.44 at the unseen $b = 4$ on the six development states (zero change 0.45, 0.53, 0.48; per-configuration median 0.55, 0.73, 0.54) and 0.22, 0.32, 0.99 at the unseen $g = 128$ across three bit-widths (zero change 1.2 to 1.3). The separable power candidate we tested is worse than zero change on the bit test and the bilinear interpolation fails outside the development corners; other amplitude-times-shape forms remain untested. On the one new state tested (1B@96k at $g = 128$, three cells) the surface is best on code and behind the source-free curves on math and QA.

Distillation pools. On the earlier protocol the reuse-count form beats the constant on the unseen pool U225 only for QA (0.706 against 1.449), with a bias that changes sign across budgets, and pool-sample variation exceeds training-seed variation (0.43 against 0.04 nats; Appendix D). Under the uniform protocol the headline joint form is compared with the frozen constant cell by cell, with paired differences and 95% intervals from a bootstrap over trajectories (Table 20). On the 270M student at the unseen pool the gain over the constant is detected on every capability (0.015, 0.011, and 0.26 nats; intervals above zero); on the 1B student it is detected for code and QA (0.040 and 0.56 nats) and the math point estimate (0.007) is not resolved by three trajectories. Against zero change the gains are 42 to 79% on every cell of both students. The pool direction is therefore covered on the development students.

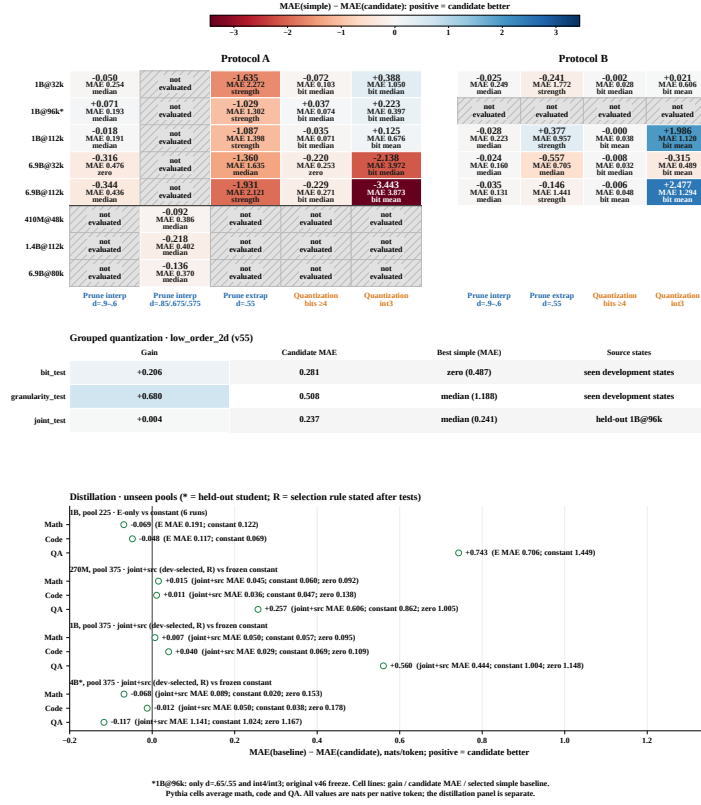


Figure 2: Generalization to unseen configurations and states. Top: improvement of the headline candidate over the strongest source-free reference (MAE difference in nats, averaged over capabilities; positive = candidate better) per new source (rows), regime (columns), and fitting protocol (panels), with the candidate MAE in each cell; hatched cells were not evaluated. Bottom: the distillation pool tests per capability against the frozen constant. All predictions were frozen before the corresponding measurement.

4.2 Transfer across Model Sizes and Training Stages

Inside the range, the frozen per-density pruning regression beat its no- D_0 variant and the per-setting median on all three capabilities at the unseen stage 96k for the 160M and 1.4B sources, and for a 1B source at steps 32k and 112k the power form, A2, and the median curve are within 0.05 nats of one another, so a new in-range size is predicted about as well by the source relation as by the development median. For a 6.9B source, five times the development range, the source-conditioned pruning forms lose to the median curve on math (0.30 against 0.12) and QA (0.91 against 0.12) and tie on code: the regression extends the panel’s size trend into QA improvements that do not occur, and the per-bit quantization regression is worse than the per-bit median at every bit-width. On 6.9B@80k the power form ties the median on math and loses on code and QA: the size boundary is soft for math and hard for QA. For distillation, on Gemma-3-4B as a held-out student (6.6 times the reference size) the frozen constant is better than the headline form at the unseen pool on every capability (intervals below zero; math 0.020 against 0.089), because the source term extends the QA reuse penalty to 2.3 nats at the largest budget where 0.40 was measured; at the development pools the headline form has the lower point error on every capability and only its QA gain clears zero (0.83 nats, interval [0.12, 1.54]; Table 20). Joint transfer to a larger student and a new pool remains unreliable.

4.3 Applicability across Capabilities and Shared Structure

Math and code are carried by the initial-state inputs on every axis, QA is not: its pruning response is over-predicted by every source-conditioned form on every new checkpoint, and its quantization error is the largest of the three on every test. Conditioning is also needed in the response shape: per-capability distillation models at the same parameter count as a shared-shape model lower the held-out error by 0.1 to 0.4 nats for the budget-plus-reuse structures (Appendix C, Fig. 3).

Shared structure, ranges, and calibration. One function family for all three methods, with method- and capability-specific parameters, holds for pruning by construction and fails for the other two arms on the same held-out splits (the separable family we tested is worse than the quantization surface, macro MAE 2.96 against 2.40; a reuse-only family is worse than the budget-plus-reuse form, 0.54 against 0.28). Within pruning one exponent shared across capabilities costs nothing (leave-one-source MAE within 0.02 nats of per-capability exponents), whereas the quantization coefficients must be capability-specific. Three ranges are kept apart (Appendix C.4): the bootstrap intervals of the pruning coefficients exclude zero for the dense-anchor term on math and code and not on QA; exponents fitted per source spread from 3.0 to 6.0 for math and 0.5 to 6.0 for QA; and 80% prediction intervals from leave-one-source residuals under-cover (78% on the 27 confirmation-panel cells, 67% on the 1B and 6.9B pair cells) at a mean full width near 1.5 nats. One measured calibration point helps or hurts by method: setting the pruning amplitude from the target’s mildest density amplifies noise through the steep exponent (leave-one-source MAE above 17 nats), whereas a separable grouped-quantization law calibrated at the single 5-bit, $g = 256$ cell predicts the unseen 4-bit configurations on held-out states with MAE 0.13, 0.12, 0.29 against 0.26, 0.64, 1.34 uncalibrated.

5 From Laws to Capability-Specific Selection

The laws of §3 are meant to be used: given an initial model, a capability, and a resource constraint, which method and setting preserves that capability best? We write the decision as

$$(m^*, \theta_m^*) = \arg \min_{m, \theta_m \in \mathcal{F}_m(B)} \widehat{L}_{m,c}(\mathbf{x}, \theta_m), \quad (6)$$

where $\mathcal{F}_m(B)$ is the set of settings of method m that meet the budget B . Every candidate is compared through the absolute loss of the deployed model (the source with its predicted response, or the smaller student after distillation); the map for one capability need not coincide with the map for another.

We validated this use retrospectively on the controlled panel with a nominal storage budget $r_{\max}$ from 0.2 to 1, using only measured configurations (211 compression cells and 9 same-stage smaller students across 17 states) and refitting every law with the target state held out. A policy with no feasible configuration at a budget returns no choice: quantization is feasible everywhere, whereas pruning-only covers 56% and distillation-only 33% of the 289 state-budget cells. Over all cells the choice made from the laws is within 0.19 (math), 0.17 (code), and 0.42 (QA) nats of the measured oracle, and within 0.03 nats for the worst of the three capabilities, picking the oracle’s method in 96%, 95%, 73%, and 94% of cells; the cheapest feasible configuration loses 4.3 to 5.2 nats. A quantization-only policy is close behind (0.21, 0.19, 0.49, and 0.04 nats), because on this storage axis quantization has the most feasible cells and dominates math and code; the maps’ contribution is therefore the connection of the laws to an explicit decision, and they show no general advantage over a strong single method on this panel. Where a smaller student exists, distillation-only is near the oracle for QA (0.06 nats; Appendix Fig. 16). Under a heuristic rule (predicted gap between the two best methods smaller than the larger of their held-out errors) about half of the cells have no clear winner; the oracle’s method lies in the reported candidate set in every such cell, and the rule carries no coverage guarantee (Appendix H.2). The maps inherit the laws’ boundaries: in-range states, nominal storage as the only cost, and unreliable QA cells.

6 Related Work

Scaling laws for compression. Existing compression laws differ in what they take as input and what they predict. Sparsity enters as a factor on the size term of a pretraining loss law (Frantar et al., 2024); precision enters through an effective-parameter count with a post-training degradation term (Kumar et al., 2025); training formats collapse onto one capacity scalar (Panferov et al., 2025); the P^2 law predicts the post-training loss of a pruned model from its size and loss before pruning, the pruning rate, and the post-training tokens (Chen et al., 2025); and a distilled student follows a broken power law in teacher loss (Busbridge et al., 2025). Their targets are aggregate losses; we follow the P^2 law in admitting the pre-compression loss and add a per-capability endpoint measured the same way for three methods, a development panel in which size and stage vary separately, and held-out tests with same-input baselines.

Task-conditioned evaluation of compression. Pruning disproportionately harms knowledge-intensive tasks (Jaiswal et al., 2024), low-bit quantization disproportionately harms reasoning (Liu et al., 2025), and selective truncation can improve specific behaviors (Sharma et al., 2024). Laws with task-level targets exist: Sengupta et al. (2025) fit pruning laws to test loss and averaged zero-shot accuracy against the pruning ratio and recovery tokens, with per-task exponents and transfer to unseen models in their revised version; Zhou et al. (2026) stratify post-training quantization laws into memorization, application, and reasoning strata for one quantizer; task-specific distillation has been mapped over dataset size, compression ratio, and supervision format (Ghita et al., 2026). Our relations use a continuous conditional loss, admit the dense capability anchor and training tokens as inputs, and treat the three methods with one protocol.

Fitting and validating empirical laws. Downstream accuracy is predictable mainly through continuous log-likelihood measurements (Schaeffer et al., 2023; 2025; Gadre et al., 2025; Bhagia et al., 2025; Lourie et al., 2025), which motivates our loss endpoint and its stated boundaries; fitted scaling-law forms are fragile to the fitting protocol (Li et al., 2025; Besiroglu et al., 2024), and repeated tokens are worth less than new tokens in data-constrained pretraining (Muennighoff et al., 2023), an effect our reuse-count coordinate measures in the distillation setting. Gradient-based importance underlies modern pruning (LeCun et al., 1989; Frantar & Alistarh, 2023; Sun et al., 2024) and enters here only as mechanism evidence (Appendix G).

7 Discussion and Conclusion

What can be predicted now is specific. Given the parameter count, the pretraining token count, and the dense capability losses of a source in the development range, the pruning form of Eq. equation 3 predicts the math and code response at any density inside the fitted range to within a few tenths of a nat. Given the same inputs, the frozen surface of Eq. equation 4 predicts the grouped quantizer at bit-widths and group sizes it was not fit on, for the states it was fit on. Given a student’s state, a linear form predicts its math transfer, and under a fixed distillation protocol a response model in budget, reuse, and a student descriptor predicts unseen data pools on the two development students. Each comes with coefficients (Appendix B) and a tested range, and §5 connects them to an explicit selection problem.

What cannot be predicted is equally specific. No source-conditioned form predicts the QA response to pruning on a new checkpoint, and no form transfers to density 0.55 or to a source five times larger. Bit-width alone supports no continuous quantization relation. Joint transfer of the distillation forms to a larger student and a new pool is unreliable, and the distillation QA gain on the primary distribution does not carry over to TriviaQA, so it is a prediction relation on the measured distribution and no general QA gain (Appendix E).

The dense capability anchor is the input to keep, the working forms are low-dimensional, and transfer across sizes is bounded by the development range. The limitations are small correlated panels, one quantizer family and one teacher, undetected code specificity on a secondary benchmark, and nominal storage as the only selection cost.

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A Experimental Details

A.1 Probes and loss computation

Each capability is measured on a fixed probe set of prompt–reference pairs drawn once, with a fixed seed, from held-out benchmark splits that are disjoint from each training and recovery source used in the paper. The primary measurement benchmarks are MATH-500 for mathematical reasoning, MBPP for code generation, and 2WikiMultihopQA for multi-hop question answering. For each capability we draw 128 probes and split them by index parity into an estimation half, used only for gradient and Fisher statistics, and a measurement half of 64 probes on which each reported loss is computed, so that mechanism statistics and evaluations do not share sampling noise. Secondary benchmarks used for the measurement checks below are GSM8K, HumanEval, and HotpotQA on the pruning panel (no training involved) and SVAMP, HumanEval, and TriviaQA where distilled models are scored, chosen to exclude the benchmarks from which the distillation traces were derived; a sample-level audit found no overlap between any measurement probe and any training pool. The measurement half is 64 probes per capability for every result in the paper except where a 512-probe re-measurement is stated.

A.2 Compression protocols

Pruning. Global magnitude pruning removes the smallest weights across all transformer matrices (attention and MLP projections; embeddings, output head, biases, and norms are excluded) to a target retained density d . The global threshold is estimated from a seeded sample of absolute weights so that the same procedure scales to 30B-parameter models; each density is applied to the same dense reference weights, which are held on the CPU for the largest models so that a single 80 GB device suffices.

Quantization. Weights are fake-quantized per output channel with symmetric round-to-nearest at bit-width b , using the integer range $[-(2^{b-1} - 1), 2^{b-1} - 1]$ and a per-channel scale set by the channel’s maximum absolute value; activations are left in bfloat16. The dense weights are restored between bit-widths. The quantizer is `fake_quantize_per_output_channel` in the released code: $q_{\max} = 2^{b-1} - 1$ is the largest positive integer, integers are clamped to $[-q_{\max}, q_{\max}]$ ($2^b - 1$ values), and the scale is the channel’s maximum absolute weight divided by $q_{\max}$. With a fixed clipping range the step

size from 4 to 5 bits shrinks by $7/15$, so the squared step ratio is $(7/15)^2 \approx 0.22$; in general the squared step ratio between adjacent bit-widths is $(q_{\max}(b)/q_{\max}(b+1))^2$, not a constant $1/4$. A squared step ratio becomes a loss ratio only under assumptions about the error distribution and the local response that we do not test, so the fixed 4^{-b} shape is one candidate among others rather than a derived reference. Nominal storage ratios are computed from bit-widths; no memory or latency was measured.

Distillation. Students are fine-tuned on black-box teacher traces (a frontier API teacher answering GSM8K, HotpotQA, and CodeAlpaca prompts; 600 traces per domain) under the full-trace recipe, with strict LoRA adapters of rank 16 and scaling 32 on the query, key, value, output, gate, up, and down projections, AdamW at learning rate 10^{-4} , an effective batch of 16 sequences, and a cosine schedule with 3% warmup. A run aborts if the adapter library is unavailable and does not fall back to full fine-tuning silently, and the actual training mode and trainable parameter count are recorded in each manifest. Token accounting distinguishes processed tokens (all input tokens seen, including repetitions), unique pool tokens, and supervised completion tokens. The data pool of U examples per domain is either the first U rows of the trace file (the historical protocol) or a random sample drawn with a pool-sampling seed that is independent of the training seed; pools are identified by the sorted hashes of their samples and pairwise overlaps are reported. The historical schedule spanned a fixed number of epochs, so its cosine horizon varied with pool size; the controlled matrix of §3.4 uses a schedule defined on the same absolute processed-token budget for each pool, and the two protocols are kept apart in fitting.

Grouped quantizer. The grouped variant (§3.3) splits each row of a weight matrix into contiguous groups of g input weights, pads the last group with zeros (which do not affect the group’s absmx), and applies the same symmetric round-to-nearest with $q_{\max} = 2^{b-1} - 1$ per group; g equal to the row width reproduces the per-channel quantizer bit-exactly (verified by a unit test). Configurations are $b \in \{3, 4, 5\}$ and $g \in \{64, 128, 256\}$; nominal storage is computed from b and g , and no memory or latency was measured.

A.3 The controlled Pythia panel

The panel uses the public Pythia checkpoints at sizes 160M, 410M, and 1.4B and pretraining steps 16,000, 64,000, and 143,000, which correspond to 33.6B, 134.2B, and 299.9B processed tokens at the published batch of 1024×2048 tokens per step. The size coordinate N_0 counts transformer-matrix parameters only (the query, key, value, output, and MLP matrices of each layer), computed from the architecture configuration, and the training-tokens coordinate is $D_0 = \text{step} \times 2,097,152$. Each checkpoint’s weight identity was verified before use: the 2.8B revisions published for different steps were found to load identical weights and were excluded, and an integrity guard now rejects any pair of nominally different checkpoints with identical measured panels. New sources added for the prospective tests (Pythia-1B at steps 32,000, 96,000, and 112,000; Pythia-6.9B at 32,000 and 112,000) were confirmed to have not entered any fit or model selection before their predictions were frozen.

The three checkpoints of the confirmation panel (410M@48k, 1.4B@112k, 6.9B@80k) were verified as distinct weights before use: their model-shard hashes differ from every other cached revision of the same size, and their dense capability vectors differ from those of the neighbouring stages.

Cohort	Sources	Interventions	Probes	Role
Heterogeneous panel	12 dense checkpoints, 5 series from 4 families: Qwen3 0.6B/1.7B/4B, Gemma-3 270M–27B, Gemma-4-31B, Muse-30B, OLMo-3 7B/32B (+2 prospective: Qwen3 8B/14B)	global magnitude pruning, 10 densities in [0.3, 0.9]; RTN int8/6/5/4/3	64 per capability, primary benchmarks	phenomena; cross-family transfer; noise screen
Controlled Pythia panel	sizes 160M/410M/1.4B $\times$ steps 16k/64k/143k (9 states); new sources 160M, 410M, 1.4B at 96k; 1B at 32k/96k/112k; 6.9B at 32k/112k	pruning: 4 dev densities 0.9–0.6, tests at 0.75/0.65/0.55; bits 8/6/4/3, test 5	same	development set; source-axis and configuration-axis tests
Distillation, controlled	the 9 Pythia states as students; fixed teacher pool (600 traces per domain); LoRA	fixed recipe	same	δ_c source transfer
Distillation, pools	Gemma-3-1B student; pools U75 and U600 (fit), U225 (test); 3 pool seeds $\times$ 2 training seeds	processed tokens T ; reuse count $E = T/D_U$	same	configuration axis (pool)
Distillation, heterogeneous	Gemma-3 270M–12B, OLMo-3-7B, Qwen3 0.6B–4B students; two API teachers; full-trace vs answer-only	recipe	same	phenomena only

Table 2: Cohorts and their roles. Counts are generated from the result files (v51_panel_tables.py); the two Qwen3 prospectives were measured at four densities only. The 2.8B Pythia checkpoints were excluded because the step revisions we downloaded loaded identical weights (hash check); nothing is claimed about other revisions.

B Fitted Models and Coefficients

The delivered pruning form (Eq. equation 3) and its standardization are in Table 4; the grouped-quantization form (Eq. equation 4) in Table 5; the per-bit source regressions, the distillation source form and pool forms, and the frozen multi-student forms are stored with their standardization constants in the register files listed in Appendix I.

capability	γ_c	$\beta_{c,0}$	$\beta_{c,\log N_0}$	β_{c,L_0}	$\beta_{c,\log D_0}$
math	3.75	+3.275	+0.319	+3.850	+0.250
code	3.05	+4.499	+0.273	+5.357	+0.347
qa	3.05	-7.409	-0.481	+5.486	+0.106

Table 4: Selected pruning form $\widehat{\Delta L_c} = (\beta_c \cdot \phi)((1 - d)/0.3)^{\gamma_c}$ with $\phi = [1, z(\log N_0), z(L_{0,c}), z(\log D_0)]$; standardization centers 20.248, 2.661, 25.614 and scales 1.407, 1.691, 0.739 for $(\log N_0, L_{0,c}, \log D_0)$ in natural units (N_0 = transformer weight-matrix parameters, D_0 = tokens). Register: v53-prune-dev/register.json.

Arm	Predictor	$P/\text{cap.}$	Varied setting; Target	Tested range	Coeff.
		dev. states	cal.		
Pruning	$A_c(\mathbf{x}) ((1 - d)/0.3)^{\gamma_c}$ (Eq. 3); same-input alternative A2 (per- d regression + interpolation)	5 (A2: 20)	d ; 17 Pythia states (160M–6.9B, 16k–143k)	0	$d \in [0.55, 0.9]$ Table 4
Quantization, fixed b	per-bit regression $\beta_{c,b} \cdot \phi(\mathbf{x})$	4 per b	new state at seen $b \in \{8, 6, 4, 3\}$; 9 dev states	0	seen b ; 5-bit register rule fails; near-zero regime un-informative
Quantization, (b, g)	$\phi(\mathbf{x}) \cdot [1, u, v, uv, u^2]$ (Eq. 4)	20	$b \in \{3, 5\} \times g \in \{64, 256\}$; 6 dev states	0	unseen $b = 4$, Table 5 $g = 128$ on dev states; new state mixed
Distillation, source axis	$\delta_c = \beta_c \cdot \phi(\mathbf{x})$ (initial student)	4	9 Pythia students, pool/budget	0	held-out stage; register math only v39
Distillation, pool axis	$a_c + b_c \log E$ (and constant, $\log T$, joint)	2	Gemma-3-1B; pools U75/U600	0	unseen pool register U225: QA v41/v50 only; multi-student test frozen

Table 3: Predictive models and their domains. Inputs are always $\mathbf{x} = (N_0, D_0, L_{0,c})$ through $\phi(\mathbf{x}) = [1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0]$; no prediction uses a measurement of the compressed target (target calibration count 0). Ranges summarize §4; coefficients and standardization constants are in Appendix B.

capability	term	β_0	$\beta_{\log N_0}$	β_{L_0}	$\beta_{\log D_0}$
math	1	+5.664	+0.569	+7.260	+0.577
math	u	-13.029	-1.306	-16.698	-1.326
math	v	+0.297	-0.026	+0.117	+0.152
math	uv	-0.162	+0.030	+0.017	-0.119
math	u^2	+7.634	+0.767	+9.786	+0.777
code	1	+5.304	+0.349	+6.539	+0.534
code	u	-12.130	-0.796	-14.945	-1.223
code	v	+0.753	-0.006	+0.619	+0.231
code	uv	-0.560	+0.011	-0.425	-0.188
code	u^2	+7.149	+0.470	+8.813	+0.720
qa	1	-9.723	-0.684	+7.403	+0.141
qa	u	+21.703	+1.561	-16.563	-0.359
qa	v	+1.825	-0.011	-1.220	+0.126
qa	uv	-1.561	+0.022	+1.028	-0.125
qa	u^2	-13.105	-0.922	+9.978	+0.190

Table 5: Grouped-quantization form $\widehat{\Delta L}_c = \sum_k \beta_{c,k} \cdot \phi(\mathbf{x}) t_k(b, g)$ with terms $t_k \in \{1, u, v, uv, u^2\}$, $u = \log_2 q_{\max}(b)$ centred at the development mean, $v = \log_2(g/128)$; standardization centers 19.565, 2.746, 25.332 and scales 1.084, 1.623, 1.095 for $(\log N_0, L_{0,c}, \log D_0)$; ridge 10^{-3} . Register: v55-quant-group/register.json.

C Alternative Functional Forms and Model Selection

C.1 Pruning: expanded development set and the selection rule

subset	candidate	math	code	QA	mean
all	power	0.328	0.355	0.798	0.494
all	A1	0.610	0.608	0.905	0.708
all	A2	0.345	0.295	0.796	0.479
all	cont	0.440	0.394	0.828	0.554
all	strength only	0.492	0.601	0.692	0.595
all	median curve	0.410	0.455	0.540	0.468
all	zero	0.812	0.852	0.602	0.755
off coarse	power	0.592	0.649	1.370	0.870
off coarse	A1	1.090	1.068	1.343	1.167
off coarse	A2	0.959	0.704	1.459	1.041
off coarse	cont	0.758	0.638	1.306	0.900
off coarse	strength only	0.909	1.077	1.163	1.050
off coarse	median curve	0.758	0.790	0.901	0.816
off coarse	zero	1.715	1.725	1.090	1.510

Table 6: Pruning development on 17 Pythia states: leave-one-source-out MAE (nats) over all held-out rows (‘all’) and over the densities off the coarse grid (0.75, 0.65, 0.55). Pre-committed rule: lowest mean over capabilities among the four source-conditioned candidates; a gap below 0.02 nats between the best two is resolved toward the continuous, zero-at- $d=1$ form with fewer parameters, which selected the power form over A2 (gap 0.015). Baselines do not select.

C.2 Grouped quantization: leave-one-state-out on six states

candidate	math	code	QA
separable	1.953	2.185	4.749
low order 2d	0.914	1.910	4.371
same input interpolation	1.230	1.889	4.333
mean	2.623	2.566	2.891
median	1.694	1.758	1.777
zero	1.821	1.929	1.796

Table 7: Grouped quantization: leave-one-state-out MAE (nats) on the 24 development cells (six states). The late-stage 160M state, which collapses at 3 bits, dominates every held-out error, so the development set does not rank the forms; all three candidates and the baselines were frozen and tested (§4.1).

Identifiability of the two-dimensional form. The development bit-widths are $b \in \{3, 5\}$, so the centred u takes the two values ± 1.16 and u^2 equals 1.35 on every development cell: the quadratic column is a multiple of the intercept column, the design of Eq. equation 4 has rank 16 for its 20 coefficients, and the ridge penalty resolves the four undetermined directions (the effective degrees of freedom are 15.9 per capability; Table 8). At the unseen $b = 4$, where $u \approx 0$ and $u^2 \approx 0$, the prediction depends on how the ridge split the fitted intercept between the two columns. A same-input design without the quadratic term spans the same development column space and gives the same development fit, yet predicts the unseen bit-width far worse (bit-test MAE 1.30, 1.39, 1.94 against 0.19, 0.21, 0.44), and so do the bit-only and bit-plus-group additive designs; at $g = 128$ all three reduced designs coincide because the group terms vanish. The frozen test result therefore stands as a result about the delivered regularized parametrization, and no curvature in bit-width is identified by the development data.

Table 8: Quantization identifiability audit, retrospective (R). The delivered V55 predictions are unchanged. The four fitted forms use the same 24 development cells per capability, pooled V55 standardization, and ridge $\lambda = 10^{-3}$ (including intercepts). The delivered 20-coefficient design has rank 16: two development bit levels make $u^2 = (u_3 + u_5)u - u_3u_5$, leaving four unidentifiable directions. Ridge chooses a unique representation but does not identify curvature; deleting u^2 also changes the induced penalty. Top: rank and effective degrees of freedom (df). Bottom: MAE in nats on the frozen bit (12 cells/capability), granularity (18), and joint 1B at 96k (3) tests, plus development leave-one-state-out (LOSO, 24), with training-only standardization refit in each fold. Median uses V55’s per-config anchors and bilinear interpolation. No control replaces a delivered prediction.

Control	Rank	df (Math)	df (Code)	df (QA)
Delivered (20)	16	15.9790	15.9801	15.9220
Without u^2 (16)	16	15.9742	15.9756	15.9045
Bit-only (8)	8	7.9871	7.9878	7.9523
Bit+group additive (12)	12	11.9797	11.9808	11.9249

Evaluation	Control	Math	Code	QA
Bit	Delivered (20)	0.1934	0.2136	0.4364
	Without u^2 (16)	1.2996	1.3863	1.9441
	Bit-only (8)	1.2996	1.3863	1.9441
	Bit+group additive (12)	1.2996	1.3863	1.9441
	V55 median	0.5533	0.7258	0.5371
	Zero	0.4499	0.5318	0.4805
Granularity	Delivered (20)	0.2189	0.3192	0.9857
	Without u^2 (16)	0.6092	0.7183	1.4759
	Bit-only (8)	0.6092	0.7183	1.4759
	Bit+group additive (12)	0.6092	0.7183	1.4759
	V55 median	1.1225	1.2219	1.2202
	Zero	1.2387	1.3457	1.2390
Joint	Delivered (20)	0.3450	0.0969	0.2689
	Without u^2 (16)	0.5271	0.2225	0.3197
	Bit-only (8)	0.5271	0.2225	0.3197
	Bit+group additive (12)	0.5271	0.2225	0.3197
	V55 median	0.1515	0.2698	0.3017
	Zero	0.2997	0.3385	0.1039
Dev LOSO	Delivered (20)	0.9135	1.9098	4.3712
	Without u^2 (16)	0.9804	1.9014	4.3649
	Bit-only (8)	1.0630	1.9014	4.3649
	Bit+group additive (12)	1.0998	1.9745	4.3768
	V55 median	1.6939	1.7582	1.7765
	Zero	1.8215	1.9291	1.7958

C.3 Distillation: descriptor forms and capability conditioning

Form	P	LOCO math	LOCO code	LOCO qa	LOSO math	LOSO code	LOSO qa
zero	0	0.152/-0.152	0.178/-0.178	1.429/+0.143	0.152/-0.152	0.178/-0.178	1.429/+0.143
constant	3	0.095/+0.000	0.103/+0.000	1.400/+0.000	0.093/+0.000	0.100/+0.000	1.358/-0.000
T-only	6	0.082/+0.000	0.072/+0.000	1.092/-0.000	0.089/+0.000	0.070/+0.000	1.075/-0.000
E-only	6	0.073/-0.003	0.055/-0.002	0.798/-0.029	0.091/+0.000	0.056/+0.000	0.949/+0.000
surface:L0	12	0.072/+0.000	0.054/-0.000	0.812/+0.001	0.091/+0.000	0.054/+0.000	0.944/-0.000
surface:logN	12	0.072/+0.000	0.054/-0.000	0.812/+0.001	0.091/+0.000	0.054/+0.000	0.944/+0.000
F1:L0	6	0.057/+0.003	0.052/+0.003	1.518/+0.662	0.152/-0.152	0.178/-0.178	1.429/+0.143
F1:logN	6	0.057/+0.003	0.052/+0.003	1.518/+0.662	0.152/-0.152	0.178/-0.178	1.429/+0.143
F2:L0	12	0.062/-0.007	0.057/-0.005	0.706/-0.069	0.152/-0.152	0.178/-0.178	1.429/+0.143
F2:logN	12	0.062/-0.007	0.057/-0.005	0.706/-0.069	0.152/-0.152	0.178/-0.178	1.429/+0.143

Table 9: Distillation development matrix (12 runs, 48 points): held-out MAE/signed bias (nats) per form and capability under leave-one-run-out (LOCO) and leave-one-student-out (LOSO). Forms: $t = \log(1 + T_c/35000)$, $e = \log(1 + E)$, $s = 1 - e^{-T_c/T_*}$; T-only $a + bt$; E-only $a + be$; surface $a + bt + ce + dz$; F1 $(a + \lambda z)e$; F2 $(a + \lambda z)s + (b + \mu z)e$, $z = \text{standardized } L_{0,c}$ (or $\log N_S$, identical up to sign with two students). Under LOSO the descriptor forms degenerate (one student in training) and fall back to zero.

Structure	P shared/per	Capability	Shared MAE	Per-cap MAE	Gain	>0.02
E-only	4/4	math	0.431	0.409	0.022	yes
E-only	4/4	code	0.417	0.412	0.006	no
E-only	4/4	qa	0.967	1.150	-0.183	no
E-only	4/4	macro	0.605	0.657	-0.052	no
T+E	5/5	math	0.439	0.327	0.111	yes
T+E	5/5	code	0.419	0.314	0.105	yes
T+E	5/5	qa	0.979	0.853	0.126	yes
T+E	5/5	macro	0.612	0.498	0.114	yes
F2	7/7	math	0.536	0.145	0.390	yes
F2	7/7	code	0.527	0.107	0.420	yes
F2	7/7	qa	1.044	0.826	0.219	yes
F2	7/7	macro	0.702	0.359	0.343	yes

Table 10: Capability conditioning at matched parameter count (leave-one-run-out MAE, nats): shared response shape with per-capability offsets versus per-capability models with the same total count; gain > 0.02 nats is a descriptive threshold.

Retrospectively, fitted on all 48 development points and scored on the 52 test points (nine unseen-pool trajectories and four held-out student trajectories at the development pools; predictions specified after the frozen matrix, origin R), the second form of Eq. equation 5 has the lowest error of every form on the 270M unseen-pool trajectories (0.022, 0.020, and 0.316 nats for math, code, and QA against 0.060, 0.047, and 0.862 for the frozen constant) and is second to the reuse-only form for QA on the 1B ones (0.478 against 0.342). The first form, which has no budget term, fails for QA on every group (1.15 to 2.26 nats). On the held-out 4B student the frozen constant remains the best predictor for math and code at the unseen pool (0.020 and 0.038), and the second form with the loss descriptor is best for QA there (0.360, with the reuse-only form at 0.367); at the development pools the two descriptor variants disagree on math (0.404 with the loss descriptor, 0.122 with the size descriptor), so the descriptor choice is not settled by these data.

C.4 Shared structure, parameter ranges, and one-point calibration

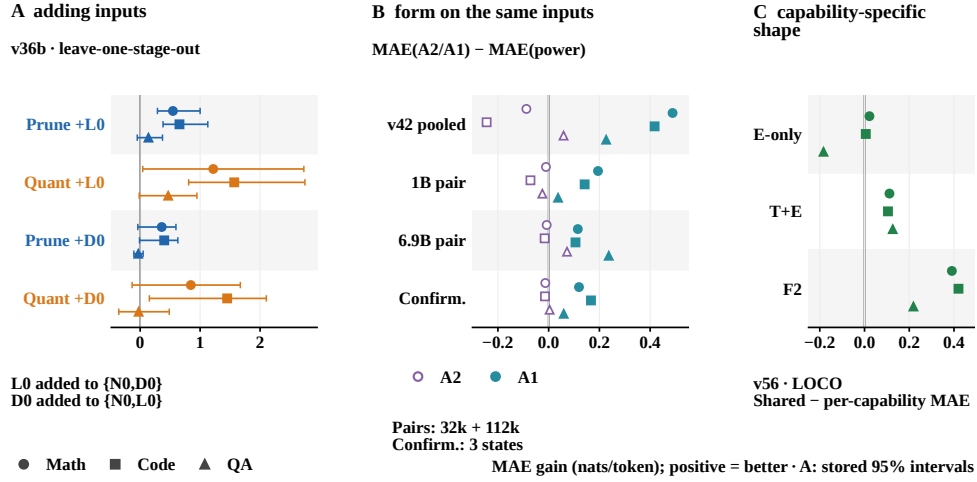


Figure 3: Which inputs and forms matter. A: improvement (nats, positive = better) from adding the dense anchor $L_{0,c}$ to (N_0, D_0) and from adding D_0 to $(N_0, L_{0,c})$ on the controlled panel, leave-one-stage-out, with 95% cluster-bootstrap intervals. B: improvement of the shared power form over the same-input alternatives A2 and A1 on every pruning test (positive = power form better). C: gain of per-capability response shapes over a shared shape at matched parameter count on the distillation development matrix (positive = per-capability better). Intervals only where computed.

Table 11 reports the shared-family test, the parameter-sharing tests, the three kinds of range (bootstrap confidence intervals of the delivered coefficients; the spread of exponents fitted separately per source; prediction intervals from leave-one-source residuals with their coverage on cells not used to build them), and the one-point calibration (K1) against the uncalibrated predictor (K0), generated by v59_shared_structure.py; K1 costs one compressed measurement per source.

Law	What shared	is	What varies	Held-out shared vs specific	MAE K0 vs MAE	K1	80% PI coverage & width
Pruning power	Power family		β_c, γ_c	0.494 vs 0.494	—	—	—
Grouped RTN	Separable family		β_c, p_c, q_c	2.962 vs 2.398	—	—	—
Distill exposure	$\log(1 + E)$ family		β_c ; inactive D_0	0.542 vs 0.275	—	—	—
Pruning sharing	One γ		β_c	0.491 vs 0.494	—	—	—
Quantization sharing	2D term vectors		Capability off-sets	3.999 vs 2.398	—	—	—
Prune math	Power family		Source amplitudes, γ	γ min/med/max 3.00/4.60/6.00	—	—	—
Prune code	Power family		Source amplitudes, γ	γ min/med/max 2.30/5.95/6.00	—	—	—
Prune qa	Power family		Source amplitudes, γ	γ min/med/max 0.50/3.40/6.00	—	—	—
Prune dev LOSO	Training γ_c		One target amplitude	—	0.604 vs 31.406	—	—
Prune confirmation	Training γ_c		One target amplitude	—	0.560 vs 0.415	77.8%/1.621 to 100.0%/3.629	to
Prune pairs: 17-source OOF	Training γ_c		One target amplitude	—	0.517 vs 0.763	—	—
Prune pairs: independent 13	Training γ_c		One target amplitude	—	0.707 vs 0.772	66.7%/1.523 to 87.5%/4.277	to
Quant dev LOSO	2D Training curve		One target amplitude	—	3.143 vs 26.177	—	—
Quant bit_test	2D Training curve		One target amplitude	—	0.747 vs 6.442	—	—
Quant granularity_test	2D Training curve		One target amplitude	—	1.751 vs 14.973	—	—
Quant joint_test	2D Training curve		One target amplitude	—	0.344 vs 0.193	—	—
Quant bit_test	sep Training curve		One target amplitude	—	1.540 vs 0.183	—	—
Quant granularity_test	sep Training curve		One target amplitude	—	2.423 vs 0.623	—	—
Quant joint_test	sep Training curve		One target amplitude	—	0.724 vs 0.306	—	—

Table 11: Shared structure and one-point calibration. MAE and PI widths are nats, pooled equally over capabilities. Family comparisons use registered dev splits: 17 pruning sources, six quantization states, 12 distillation runs. Specific forms: v53 power, v55 low-order 2D, v56 F2 with L_0 . Distillation D_0 is unavailable (inactive column). Gamma ranges describe within-source fits. Bootstrap parameter CIs are in the accompanying summary. K1 costs one compressed measurement per source; scores exclude that cell. PI entries give coverage/full width for K0 then K1, from per-capability absolute LOSO residual quantiles. Confirmation excludes three new sources; independent pair evaluation excludes all four pair sources from fitting and its 13-source interval bank. Quantization tests omit the target source from fitting; 2D K1 scales its predicted curve and sep K1 calibrates the separable amplitude.

D Complete Prediction Results

Tables 13–15 give every registered test in three lines (candidate, strongest same-input baseline, strongest simple baseline) with improvements and provenance codes; Table 16 the

per-stage, per-protocol results of the Pythia-1B and 6.9B pairs; Tables 17 and 18 the confirmation panel and the grouped-quantization tests; Fig. 4 the paired improvements behind Fig. 3.

Test	unseen	origin	math	code	QA
Source axis: seen setting, unseen source state					
prune: fixed d , 3×3 , leave-one-stage-out	stage	RFRA	0.253/0.413 _{med}	0.292/0.516 _{med}	0.806/0.500 _o
quant: fixed b , 3×3 , leave-one-stage-out	stage	RFRA	1.144/1.410 _{med}	0.626/1.553 _{med}	1.989/1.473 _{med}
prune: stage 96k (160M, 1.4B), all d	stage	PFRA	0.090/0.343 _{nD}	0.128/0.179 _{nD}	0.426/0.438 _{nD}
quant: new stage 96k (160M/410M/1.4B), $b \geq 4$	stage	P-RA	0.029/0.084 _{pbm}	0.104/0.094 _{pbm}	0.173/0.134 _{pbm}
quant: new stage 96k (160M/410M/1.4B), int3	stage	P-RA	1.488/1.934 _{pbm}	1.963/0.843 _{pbm}	2.046/1.727 _{pbm}
distill: δ_c , 3×3 LoRA, leave-one-stage-out	stage	RFRA	0.014/0.026 _{nD}	0.057/0.044 _{nD}	0.263/0.159 _c
Configuration axis: unseen setting, size, stage, or pool					
prune: d 0.65/0.55, dev sources + 410M@96k	d	PRFA	0.371/0.283 _{A2}	0.524/0.279 _{A2}	0.759/0.818 _{A2}
prune: 1B@96k, $d = 0.65$	size, stage, d	PFFA	0.020/0.009 _{A1}	0.092/0.031 _{A1}	0.466/0.380 _{A2}
prune: 1B@96k, $d = 0.55$	size, stage, d	PFFA	0.930/0.273 _{so}	1.550/0.018 _{so}	1.426/0.530 _{so}
quant: 1B@96k, int4	size, stage	PFFA	0.116/0.068 _{pbm}	0.046/0.012 _{pbm}	0.059/0.054 _{nD}
quant: 1B@96k, int3	size, stage	PFFA	0.690/1.414 _{pbm}	0.318/0.001 _{pbm}	0.182/0.446 _{pbm}
prune: 1B@32k+112k, d 0.9–0.6	size, stage	PFFA	0.215/0.169 _{so}	0.191/0.119 _{A2}	0.261/0.227 _{med}
prune: 1B@32k+112k, $d = 0.55$	size, stage, d	PFFA	1.790/0.939 _{so}	1.488/0.089 _{so}	2.225/0.393 _{so}
quant: 1B@32k+112k, $b \geq 4$ (5-bit rule incl.)	size, stage, b	PFFA	0.064/0.020 _{pbm}	0.041/0.028 _{pbm}	0.157/0.052 _{pbm}
quant: 1B@32k+112k, int3	size, stage	PFFA	0.256/1.466 _{pba}	0.362/1.537 _{pba}	1.972/1.000 _{pbm}
prune: 6.9B@32k+112k, d 0.9–0.6	size $\uparrow$, stage	PFFA	0.299/0.120 _{med}	0.160/0.144 _{A2}	0.909/0.120 _{med}
prune: 6.9B@32k+112k, $d = 0.55$	size $\uparrow$, stage, d	PFFA	0.615/0.618 _{A2}	0.547/0.535 _{A2}	4.472/0.904 _{so}
quant: 6.9B@32k+112k, $b \geq 4$	size $\uparrow$, stage, b	PFFA	0.181/0.036 _{pbm}	0.177/0.043 _{pbm}	0.428/0.034 _o
quant: 6.9B@32k+112k, int3	size $\uparrow$, stage	PFFA	3.233/2.046 _{pba}	3.229/1.805 _{pba}	5.306/1.744 _{pba}
distill: unseen pool U225 (Gemma-3-1B)	pool	PFFA	0.191/0.122 _c	0.117/0.069 _c	0.706/0.710 _{TE}
KD 270M, pool 375	pool	P/F/F/A	0.045/0.060 _c	0.036/0.047 _c	0.606/0.862 _c
KD 1B, pool 375	pool	P/F/F/A	0.050/0.057 _c	0.029/0.069 _c	0.444/1.004 _c
KD 4B, pool 375	student, pool	P/F/F/A	0.089/0.020 _c	0.050/0.038 _c	1.141/1.024 _c

Table 12: Prediction tests, all three arms, complete list (MAE in nats/token per capability, candidate / strongest baseline, bold = better). The candidate is the source-conditioned predictor of each arm (configuration-indicator regression on the source axis and for quantization; shared power form for pruning on the configuration axis; E -only form for the U225 distillation pool test; joint form with a source term for the multi-student U375 tests). For U375, the headline is the frozen form with the lowest per-capability development MAE; this selection rule was stated after the tests (R). Each U375 group has 12 points per capability; “student” marks the held-out Gemma-3-4B. The baseline shown is the stronger of the same-input and the simple baseline per capability, named by code: nD = no- D_0 variant, med = per-configuration median or median development curve, 0 = zero change, so = strength-only curve, A1 = $\gamma=1$ power form, A2 = per-density regression with fixed interpolation, ct = continuous two-term form, pbm/pba = per-bit median/mean, c = constant, TE = joint $T+E$ form. “unseen” lists what the test holds out (size $\uparrow$ = five times beyond the development range). Origin code, four letters: prediction (P frozen before measurement regardless of outcome, R retrospective, including leave-one-out), same-input baseline, simple baseline (F pre-specified with the candidate, R computed after the test, – none), selection (A: all pre-specified forms reported; S: per-capability choice made after the test). Improvements, both baselines separately, and bootstrap intervals are in Appendix Tables 13–15. Generated by v52_prediction_tables.py.

Test; unseen axis; [origin]	Line	math	code	QA
pruning: source transfer at fixed d : 3×3 Pythia panel, leave-one-step-out held-out pretraining stage (in-range) $[R \cdot F \cdot R \cdot A]$	cand.: config-indicator OLS	0.253 (+0.16)	0.292 (+0.22)*	0.806 (-0.31)
	same-input: no- D_0	0.613	0.693	0.774
	simple: per-config	0.413	0.516	0.500
	median; zero change			
quantization: source transfer at fixed b : 3×3 Pythia panel, leave-one-step-out held-out pretraining stage (in-range) $[R \cdot F \cdot R \cdot A]$	cand.: config-indicator OLS	1.144 (+0.27)	0.626 (+0.93)*	1.989 (-0.52)
	same-input: no- D_0	1.989	2.077	1.964
	simple: per-config	1.410	1.553	1.473
	median			
pruning: new stage 96k on 160M and 1.4B, all measured d stage 96k (interpolation between 64k and 143k) $[P \cdot F \cdot R \cdot A]$	cand.: config-indicator OLS	0.090 (+0.25)	0.128 (+0.05)	0.426 (+0.01)
	same-input: no- D_0	0.343	0.179	0.438
	simple: per-config	0.367	0.438	0.832
	median			
quantization: new stage 96k on 160M, 410M, 1.4B; $b \geq 4$ (near-zero regime) stage 96k (interpolation) $[P \cdot - \cdot R \cdot A]$	cand.: config-indicator OLS	0.029 (+0.06)	0.104 (-0.01)	0.173 (-0.04)
	same-input: none available	—	—	—
	simple: per-bit	0.084	0.094	0.134
	median			
quantization: new stage 96k on 160M, 410M, 1.4B; int3 (collapse regime) stage 96k (interpolation) $[P \cdot - \cdot R \cdot A]$	cand.: config-indicator OLS	1.488 (+0.45)	1.963 (-1.12)	2.046 (-0.32)
	same-input: none available	—	—	—
	simple: per-bit	1.934	0.843	1.727
	median			
distillation: δ_c source transfer: 3×3 LoRA panel, leave-one-step-out held-out pretraining stage (in-range) $[R \cdot F \cdot R \cdot A]$	cand.: linear $\{N_0, L_0, D_0\}$	0.014 (+0.01)	0.057 (-0.01)	0.263 (-0.10)
	same-input: no- D_0	0.026	0.044	0.265
	simple: constant (mean); constant (median)	0.031	0.061	0.159

Table 13: Source axis: does the pre-compression source state predict the response at a fixed intervention setting? MAE in nats/token per capability; bold = smallest in the group; the number in parentheses is the improvement of the candidate over the strongest listed baseline (positive = candidate better); * = bootstrap 95% interval of that improvement excludes zero (computed only for the leave-one-out panels). Origin code: prediction origin (P frozen before measurement regardless of outcome, R retrospective, including leave-one-out) · same-input baseline origin · simple baseline origin (F pre-specified with the candidate, R computed after the test, — none) · selection (A: all pre-specified forms reported; S: per-capability choice made after the test). Where several same-input or simple baselines exist, the strongest per capability is shown and named; that choice is retrospective. Config-indicator OLS uses $\{N_0, L_0, D_0\}$ with one coefficient set per configuration. Generated by v52_prediction_tables.py.

Test; unseen axis; [origin]	Line	math	code	QA
pruning: unseen d 0.65 and 0.55, dev sources + 410M@96k densities 0.65 (interp.) and 0.55 (extrap.) pooled [P · R · F · A]	cand.: shared power form	0.371 (-0.09)	0.524 (-0.24)	0.759 (+0.06)
	same-input: A2 (per- d regr.)	0.283	0.279	0.818
	simple: strength-only	1.369	1.635	1.608
pruning: new source 1B@96k, $d = 0.65$ (interp.) in-range size; stage 96k [P · F · F · A]	cand.: shared power form	0.020 (-0.01)	0.092 (-0.06)	0.466 (-0.09)
	same-input: A1 ($\gamma=1$); A2 (per- d regr.)	0.009	0.031	0.380
	simple: median curve	0.147	0.076	0.569
pruning: new source 1B@96k, $d = 0.55$ (extrap.) in-range size; stage 96k [P · F · F · A]	cand.: shared power form	0.930 (-0.66)	1.550 (-1.53)	1.426 (-0.90)
	same-input: A2 (per- d regr.)	1.051	1.358	1.009
	simple: strength-only	0.273	0.018	0.530
pruning: new source 1B@32k+112k, d 0.9–0.6 (interp.) in-range size; stages 32k, 112k [P · F · F · A]	cand.: shared power form	0.215 (-0.05)	0.191 (-0.07)	0.261 (-0.03)
	same-input: A2 (per- d regr.)	0.205	0.119	0.237
	simple: median curve; strength-only	0.169	0.128	0.227
pruning: new source 1B@32k+112k, $d = 0.55$ (extrap.) in-range size; stages 32k, 112k [P · F · F · A]	cand.: shared power form	1.790 (-0.85)	1.488 (-1.40)	2.225 (-1.83)
	same-input: A1 ($\gamma=1$); A2 (per- d regr.)	1.896	1.386	1.744
	simple: strength-only	0.939	0.089	0.393
pruning: new source 6.9B@32k+112k, d 0.9–0.6 (interp.) size $5\times$ beyond dev range; stages 32k, 112k [P · F · F · A]	cand.: shared power form	0.299 (-0.18)	0.160 (-0.02)	0.909 (-0.79)
	same-input: A2 (per- d regr.)	0.292	0.144	0.982
	simple: median curve	0.120	0.144	0.120
pruning: new source 6.9B@32k+112k, $d = 0.55$ (extrap.) size $5\times$ beyond dev range; stages 32k, 112k [P · F · F · A]	cand.: shared power form	0.615 (+0.00)	0.547 (-0.01)	4.472 (-3.57)
	same-input: A1 ($\gamma=1$); A2 (per- d regr.)	0.618	0.535	2.850
	simple: median curve; strength-only	0.703	0.712	0.904

Table 14: Configuration axis, pruning: does a relation fit at seen densities and sources predict unseen densities, sizes and stages? Protocol A (full development panel) for the two P1-v2 pairs and the single-stage 1B@96k prospective; protocol B in Table 16. Format, bold, improvement in parentheses, and origin code as in Table 13.

Test; unseen axis; [origin]	Line	math	code	QA
quantization: new source 1B@96k, int4 in-range size; stage 96k; seen bit $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.116 (-0.05)	0.046 (-0.03)	0.059 (-0.00)
	same-input: no- D_0	0.184	0.394	0.054
	simple: per-bit median	0.068	0.012	0.253
quantization: new source 1B@96k, int3 in-range size; stage 96k; seen bit $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.690 (+0.72)	0.318 (-0.32)	0.182 (+0.26)
	same-input: no- D_0	2.324	3.423	1.635
	simple: per-bit median	1.414	0.001	0.446
quantization: new source 1B@32k+112k, $b \geq 4$ (incl. 5-bit rule) in-range size; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.064 (-0.04)	0.041 (-0.01)	0.157 (-0.10)
	same-input: no- D_0	0.118	0.156	0.153
	simple: per-bit median	0.020	0.028	0.052
quantization: new source 1B@32k+112k, int3 in-range size; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.256 (+1.21)	0.362 (+1.18)	1.972 (-0.97)
	same-input: no- D_0	3.218	3.565	2.897
	simple: per-bit mean; per-bit median	1.466	1.537	1.000
quantization: new source 6.9B@32k+112k, $b \geq 4$ (incl. 5-bit rule) size $5\times$ beyond dev range; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.181 (-0.14)	0.177 (-0.13)	0.428 (-0.39)
	same-input: no- D_0	0.169	0.154	0.426
	simple: per-bit median; zero change	0.036	0.043	0.034
quantization: new source 6.9B@32k+112k, int3 size $5\times$ beyond dev range; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	3.233 (-1.19)	3.229 (-1.42)	5.306 (-3.56)
	same-input: no- D_0	4.199	3.176	5.605
	simple: per-bit mean	2.046	1.805	1.744
distillation: unseen pool U225 from endpoint pools U75/U600 (Gemma-3-1B) pool (reuse count in-range; sampling protocol differs) $[P \cdot F \cdot F \cdot A]$	cand.: E -only (reuse count)	0.191 (-0.07)	0.117 (-0.05)	0.706 (+0.00)
	same-input: $T+E$ joint	0.215	0.145	0.710
	simple: constant	0.122	0.069	1.449

D.1 Pythia-1B and 6.9B pairs under two protocols

The second frozen round (Table 16) added seven pruning densities (0.9, 0.8, 0.75, 0.7, 0.65 and 0.6 as interpolation, 0.55 as extrapolation) and five bit-widths (8, 6, 5, 4, 3) on Pythia-1B at steps 32k and 112k. Two fitting protocols were frozen at the same time. Protocol A fits each candidate on the full nine-state development panel. Protocol B fits on development states at step 64k or below, so that its 112k rows test a training-stage extrapolation on top of the size change. The 5-bit prediction of the class-indicator candidate is a fixed rule, the average of its committed 4- and 6-bit predictions, chosen so that no 5-bit measurement enters any fit. Candidate names follow §3.2: power is the shared power form, A2 the per-density source regression with a fixed interpolation rule, A1 its $\gamma = 1$ variant, cont a continuous two-term polynomial in $1 - d$, and the three source-free baselines are strength-only, the median development curve, and zero change. For quantization, full uses $\{N_0, L_0, D_0\}$, no-D0 drops the training-stage term, and mean and median are per-bit constants from the development panel.

Per capability, at 112k under protocol A the source regression reaches MAE 0.111 (math), 0.091 (code) and 0.135 (QA); the strongest source-free alternative is strength-only for math (0.117) and the median curve for code (0.107) and QA (0.261). At 32k the QA error of the source regression rises to 0.339 while the median curve stays at 0.193, which accounts for most of the aggregate gap. Under protocol B at 112k the power and A2 forms are ahead of strength-only at $d = 0.55$ (0.96 and 1.00 against 1.33), but with errors near one nat the ordering carries little practical value. The pair cost 22 A100 minutes.

The 6.9B pair, a size extrapolation of about five times, was measured under the same freeze (36 A100 minutes, reference model on CPU). Its dense anchors are math 1.27, code 1.30 and QA 5.11 at 32k, and 1.11, 1.21 and 5.09 at 112k. Per capability under protocol A at 112k, the source regression reaches 0.23 (math), 0.19 (code) and 0.96 (QA) against 0.05, 0.07 and 0.15 for the median curve. Under protocol B the source forms lead on code only (A2 0.058 and 0.083 at the two stages against 0.208 and 0.074 for the median curve) and trail on QA (0.32 and 0.20 against 0.08 and 0.15). At $d = 0.55$ the source-free curves win under both protocols, with source-conditioned errors between 0.55 and 2.2 nats. For quantization under protocol A the class-indicator candidate is worst or near worst above 3 bits (0.25 and 0.27 against 0.03 and 0.04 for the per-bit median) and overshoots by about four nats at 3 bits. Protocol B compresses the differences above 3 bits to the 0.02 to 0.05 range and leaves the candidate ahead at 3 bits only for the late stage (1.29 against 3.8 to 4.1).

The multi-student distillation test of the uniform protocol (§3.4) is given in Table 19; the frozen predictions of all eight forms for every test point are in the mirrored compare file. Two pre-specified training-seed repeats (Gemma-3-270M at $U=75$ and Gemma-3-1B at $U=450$, data seed 11) reproduce the four checkpoints of their originals within 0.017 nats for math and 0.030 for code; for QA the difference grows with the reuse count, from 0.004 to 0.019 at the first two checkpoints to 0.089 and 0.186 at the last, so training-seed noise is well below the law errors except for QA at high reuse.

Table 19: Multi-student distillation test on the uniform absolute-exposure protocol. Eight forms per capability were frozen from the twelve development runs (Gemma-3-270M and 1B, $U \in \{75, 450\}$, three data seeds) before any test run; the headline column is the frozen form with the lowest development error at freeze time (joint with a source term $k \log(N_S/N_{\text{ref}})$ for every capability; this selection rule was stated after the tests). MAE in nats over all test points of the row (three data seeds times four budgets per trajectory), the frozen per-capability constant (the same-input baseline of this arm), the zero-change baseline, and the best of the eight frozen forms chosen after the fact (R). Gemma-3-4B is a held-out student whose $\log N_S$ lies 1.9 above the reference against a development span of ± 0.55 .

Student, pools	Cap.	Dev-selected	Constant	Zero	Best frozen (R)
Gemma-3-270M, unseen pool ($U=375$)	Math	0.045	0.060	0.092	constant+src (0.031)
	Code	0.036	0.047	0.138	E (0.031)
	QA	0.606	0.862	1.005	joint (0.544)
Gemma-3-1B, unseen pool ($U=375$)	Math	0.050	0.057	0.095	E (0.026)
	Code	0.029	0.069	0.109	E (0.018)
	QA	0.444	1.004	1.148	joint (0.301)
Gemma-3-4B, unseen pool ($U=375$)	Math	0.089	0.020	0.153	constant (0.020)
	Code	0.050	0.038	0.178	T (0.033)
	QA	1.141	1.024	1.167	joint (0.272)
Gemma-3-4B, dev pools ($U=75, 450$), held-out student	Math	0.106	0.156	0.294	joint+src (0.106)
	Code	0.048	0.098	0.230	joint+src (0.048)
	QA	0.958	1.790	1.862	joint (0.759)

Table 20: Paired distillation errors for the frozen joint+src candidate. MAEs and paired differences are in nats; positive differences and relative improvements favor the candidate. Differences are point means of $|e_{\text{baseline}}| - |e_{\text{candidate}}|$; relative improvement is $100(\text{MAE}_{\text{baseline}} - \text{MAE}_{\text{candidate}})/\text{MAE}_{\text{baseline}}$. Brackets show 95% percentile intervals from 5,000 paired trajectory bootstrap resamples (seed 0); a cluster is student $\times$ pool, with all budgets resampled together. The same draws are used for both baselines and all capabilities within each student/pool-role group, recomputing each ratio per resample. Test rows use 12 points from 3 trajectories; held-out-student development rows use 16 points from 4 trajectories. Intervals have limited resolution with so few trajectories.

Student	Cap.	Candidate MAE	Frozen constant baseline			Zero baseline		
			MAE	Diff. [CI]	Rel. % [CI]	MAE	Diff. [CI]	Rel. % [CI]
Test pools								
270M	Math	0.045	0.060	0.015	25.2	0.046	50.5	
				[0.002, 0.027]	[3.9, 39.9]	[0.043, 0.049]	[49.2, 51.5]	
270M	Code	0.036	0.047	0.011	22.7	0.102	73.7	
				[0.006, 0.017]	[14.0, 35.0]	[0.097, 0.109]	[72.4, 75.2]	
270M	QA	0.606	0.862	0.257	29.8	0.400	39.8	
				[0.173, 0.376]	[21.5, 39.3]	[0.316, 0.519]	[33.4, 47.2]	
1B	Math	0.050	0.057	0.007	11.8	0.045	47.0	
				[−0.001, 0.015]	[−1.6, 23.4]	[0.040, 0.052]	[42.8, 50.8]	
1B	Code	0.029	0.069	0.040	58.0	0.080	73.5	
				[0.032, 0.044]	[52.5, 64.0]	[0.068, 0.089]	[67.3, 77.0]	
1B	QA	0.444	1.004	0.560	55.8	0.704	61.3	
				[0.555, 0.566]	[53.9, 58.0]	[0.698, 0.709]	[59.5, 63.4]	
4B	Math	0.089	0.020	−0.068	−333.8	0.064	42.0	
				[−0.070, −0.066]	[−486.0, −265.6]	[0.043, 0.080]	[31.1, 48.8]	
4B	Code	0.050	0.038	−0.012	−31.5	0.127	71.6	
				[−0.029, −0.003]	[−71.8, −7.9]	[0.117, 0.133]	[65.4, 76.9]	
4B	QA	1.141	1.024	−0.117	−11.4	0.026	2.2	
				[−0.119, −0.115]	[−11.8, −11.1]	[0.024, 0.028]	[2.1, 2.4]	
Development pools, held-out 4B student								
4B	Math	0.106	0.156	0.049	31.6	0.187	63.8	
				[−0.068, 0.166]	[−573.4, 55.6]	[0.069, 0.306]	[46.4, 69.7]	
4B	Code	0.048	0.098	0.050	50.7	0.182	79.1	
				[−0.019, 0.118]	[−58.9, 72.0]	[0.103, 0.261]	[67.2, 87.0]	
4B	QA	0.958	1.790	0.832	46.5	0.904	48.5	
				[0.125, 1.540]	[11.7, 62.4]	[0.268, 1.540]	[22.1, 63.1]	

Test; unseen axis; [origin]	Line	math	code	QA
distillation (uniform absolute-exposure protocol): multi-student distillation, unseen pool U375, Gemma-3-270M pool [P/F/F/A]	cand.: joint form with source term (dev-selected, R)	0.045 (+0.02)	0.036 (+0.01)	0.606 (+0.26)
	same-input: constant	0.060	0.047	0.862
	simple: zero change	0.092	0.138	1.005
distillation (uniform absolute-exposure protocol): multi-student distillation, unseen pool U375, Gemma-3-1B pool [P/F/F/A]	cand.: joint form with source term (dev-selected, R)	0.050 (+0.01)	0.029 (+0.04)	0.444 (+0.56)
	same-input: constant	0.057	0.069	1.004
	simple: zero change	0.095	0.109	1.148
distillation (uniform absolute-exposure protocol): multi-student distillation, unseen pool U375, held-out student Gemma-3-4B pool; held-out student [P/F/F/A]	cand.: joint form with source term (dev-selected, R)	0.089 (-0.07)	0.050 (-0.01)	1.141 (-0.12)
	same-input: constant	0.020	0.038	1.024
	simple: zero change	0.153	0.178	1.167

Table 15: Configuration axis, quantization and distillation (continued from the preceding table block): unseen sizes and stages at seen bit-widths (protocol A; the 5-bit prediction is a fixed interpolation rule), and unseen distillation pools. The U375 headline is the frozen form with the lowest per-capability development MAE (joint with a source term for every capability); this selection rule was stated after the tests (R). Each U375 group has 12 points per capability; Gemma-3-4B is a held-out student. Format, bold, improvement in parentheses, and origin code as in Table 13.

Source	Prot.	Regime	power	A2	A1	cont	str.-only	median	zero	best
1b@32000	A	prune interp d0.9-0.6	0.254	0.261	0.393	0.275	0.363	0.204	0.412	median-curve
1b@32000	A	prune extrap d0.55	2.272	2.130	2.394	2.221	0.637	1.509	2.434	strength-only
1b@32000	B	prune interp d0.9-0.6	0.249	0.218	0.321	0.239	0.236	0.224	0.412	A2
1b@32000	B	prune extrap d0.55	1.772	1.777	2.151	1.863	1.531	1.645	2.434	strength-only
1b@112000	A	prune interp d0.9-0.6	0.191	0.112	0.300	0.175	0.365	0.173	0.400	A2
1b@112000	A	prune extrap d0.55	1.398	1.235	1.838	1.356	0.311	1.311	2.237	strength-only
1b@112000	B	prune interp d0.9-0.6	0.223	0.177	0.355	0.235	0.219	0.195	0.400	A2
1b@112000	B	prune extrap d0.55	0.957	0.995	1.685	1.127	1.333	1.447	2.237	power
6.9b@32000	A	prune interp d0.9-0.6	0.476	0.482	0.608	0.501	0.483	0.165	0.160	zero
6.9b@32000	A	prune extrap d0.55	1.635	1.423	0.939	1.457	1.493	0.275	0.889	median-curve
6.9b@32000	B	prune interp d0.9-0.6	0.160	0.166	0.249	0.187	0.163	0.136	0.160	median-curve
6.9b@32000	B	prune extrap d0.55	0.705	0.549	0.677	0.630	0.252	0.148	0.889	median-curve
6.9b@112000	A	prune interp d0.9-0.6	0.436	0.463	0.610	0.479	0.376	0.092	0.294	median-curve
6.9b@112000	A	prune extrap d0.55	2.121	1.939	2.224	2.061	0.191	1.272	2.198	strength-only
6.9b@112000	B	prune interp d0.9-0.6	0.131	0.106	0.235	0.140	0.111	0.095	0.294	median-curve
6.9b@112000	B	prune extrap d0.55	1.441	1.387	1.892	1.525	1.294	1.408	2.198	strength-only
Source	Prot.	Regime	full	no-D0	mean	median	zero	best		
1b@32000	A	quant ≥ 4 -bit	0.103	0.174	0.197	0.031	0.046	per-bit-median		
1b@32000	A	quant int3	1.050	2.648	2.903	1.438	3.108	full		
1b@32000	A	quant int5 (rule)	0.179	0.272	0.254	0.044	0.017	zero		
1b@32000	B	quant ≥ 4 -bit	0.028	0.032	0.027	0.026	0.046	per-bit-median		
1b@32000	B	quant int3	0.606	1.004	0.627	0.803	3.108	full		
1b@32000	B	quant int5 (rule)	0.025	0.013	0.035	0.034	0.017	noD0		
1b@112000	A	quant ≥ 4 -bit	0.071	0.111	0.183	0.036	0.063	per-bit-median		
1b@112000	A	quant int3	0.676	3.805	0.802	1.231	5.777	full		
1b@112000	A	quant int5 (rule)	0.151	0.138	0.251	0.035	0.025	zero		
1b@112000	B	quant ≥ 4 -bit	0.038	0.042	0.038	0.039	0.063	per-bit-mean		
1b@112000	B	quant int3	1.120	1.981	3.106	3.472	5.777	full		
1b@112000	B	quant int5 (rule)	0.046	0.019	0.027	0.025	0.025	noD0		
6.9b@32000	A	quant ≥ 4 -bit	0.253	0.317	0.190	0.034	0.033	zero		
6.9b@32000	A	quant int3	3.973	5.103	3.300	1.835	2.711	per-bit-median		
6.9b@32000	A	quant int5 (rule)	0.373	0.457	0.237	0.067	0.029	zero		
6.9b@32000	B	quant ≥ 4 -bit	0.033	0.050	0.029	0.025	0.033	per-bit-median		
6.9b@32000	B	quant int3	0.489	1.792	0.174	0.407	2.711	per-bit-mean		
6.9b@32000	B	quant int5 (rule)	0.046	0.052	0.058	0.057	0.029	zero		
6.9b@112000	A	quant ≥ 4 -bit	0.271	0.183	0.155	0.042	0.061	per-bit-median		
6.9b@112000	A	quant int3	3.873	3.550	0.430	1.896	6.442	per-bit-mean		
6.9b@112000	A	quant int5 (rule)	0.402	0.247	0.253	0.033	0.027	zero		
6.9b@112000	B	quant ≥ 4 -bit	0.048	0.060	0.042	0.045	0.061	per-bit-mean		
6.9b@112000	B	quant int3	1.294	4.033	3.770	4.137	6.442	full		
6.9b@112000	B	quant int5 (rule)	0.043	0.025	0.025	0.023	0.027	per-bit-median		

Table 16: P1-v2 new-source frozen prospectives (MAE over three capabilities, nats/token). Protocol A fits on the full nine-state development panel; protocol B on step ≤ 64 k only, so its 112k rows are training-stage extrapolations. 1B sources are in-range sizes and 6.9B is a $\sim 5\times$ size extrapolation. Predictions were committed before measurement; per-capability and per-point values are in the results ledger.

Candidate	P	test MAE			signed bias (pred. - obs.)		
		math	code	QA	math	code	QA
power (selected)	5	0.243	0.236	0.678	+0.216	+0.236	+0.382
A2	20	0.230	0.222	0.682	+0.222	+0.212	+0.293
A1	4	0.363	0.403	0.738	+0.210	+0.206	+0.304
cont	8	0.313	0.305	0.746	+0.116	+0.107	+0.249
strength only	2	0.450	0.488	0.579	+0.443	+0.486	+0.579
median curve	0	0.277	0.214	0.221	+0.277	+0.214	+0.186
zero	0	0.522	0.530	0.185	-0.522	-0.530	+0.185

Table 17: Round-3 pruning deliverable. Forms: power $(\beta_c \cdot \phi)((1-d)/0.3)^{\gamma_c}$; A2 = per-density OLS on ϕ at five anchors with linear interpolation; A1 = power form with $\gamma_c=1$; cont = $(\beta_c \cdot \phi)s + (\zeta_c \cdot \phi)s^2$, $s=1-d$; strength-only $A((1-d)/0.3)^{\gamma}$, the median development curve, and zero change are source-free. Inputs $\phi = [1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0]$ (standardized on the development set); development set = 17 Pythia states (160M–6.9B, steps 16k–143k, densities ≥ 0.55); the selected form was chosen by a pre-committed leave-one-source-out rule (tie within 0.02 nats resolved toward the continuous, zero-at- $d=1$ form with fewer parameters). Test = three new checkpoints (410M@48k, 1.4B@112k, 6.9B@80k) at $d \in \{0.85, 0.675, 0.575\}$, predictions committed before measurement; MAE and mean signed error (prediction minus observation) over nine cells per capability. Coefficients in Appendix Table 4. Gain of the selected form vs A2 (same input): math -0.013, code -0.014, qa +0.004; vs the strongest source-free curve: math +0.034, code -0.023, qa -0.494. Generated by v57_round3_tables.py.

Candidate	P	bit test (12)			granularity test (18)			joint, 1B@96k (3)		
		math	code	QA	math	code	QA	math	code	QA
low-order 2D	20	0.193	0.214	0.436	0.219	0.319	0.986	0.345	0.097	0.269
separable	6	0.767	0.798	1.427	0.581	0.643	1.612	0.586	0.380	0.679
same input interpolation	16	1.296	1.381	1.942	0.606	0.711	1.497	0.538	0.241	0.392
median	0	0.553	0.726	0.537	1.123	1.222	1.220	0.152	0.270	0.302
mean	0	1.416	1.556	1.495	1.857	1.906	2.019	1.491	1.558	1.813
zero	0	0.450	0.532	0.480	1.239	1.346	1.239	0.300	0.339	0.104

Table 18: Round-3 grouped-quantization deliverable (MAE, nats). Forms: low-order 2D $= \phi \cdot [1, u, v, uv, u^2]$ with $u = \log_2 q_{\max}$ (centred) and $v = \log_2(g/128)$, one 4-vector of state coefficients per term; separable $= (\beta_c \cdot \phi) q_{\max}^{-p_c}(g/128)^{q_c}$; same-input interpolation = bilinear between the four development configurations; median and mean = per-configuration statistics interpolated the same way (source-free). Same inputs ϕ as Table 17; development = six Pythia states (160M/410M/1.4B at steps 16k and 143k) $\times b \in \{3, 5\} \times g \in \{64, 256\}$ under symmetric round-to-nearest with contiguous input groups of g weights; all predictions committed before the tests. Bit test: unseen $b=4$ at $g \in \{64, 256\}$ on the dev states; granularity test: unseen $g=128$ at $b \in \{3, 4, 5\}$ on the dev states; joint test: new state 1B@96k at $g=128$. All candidates and baselines are reported on every test (pre-committed rule; no selection by outcome). Signed bias of the 2D form (math/code/QA): bit +0.108/+0.060/+0.077; granularity +0.162/+0.137/+0.129; joint +0.345/+0.097/+0.250. Generated by v57_round3_tables.py.

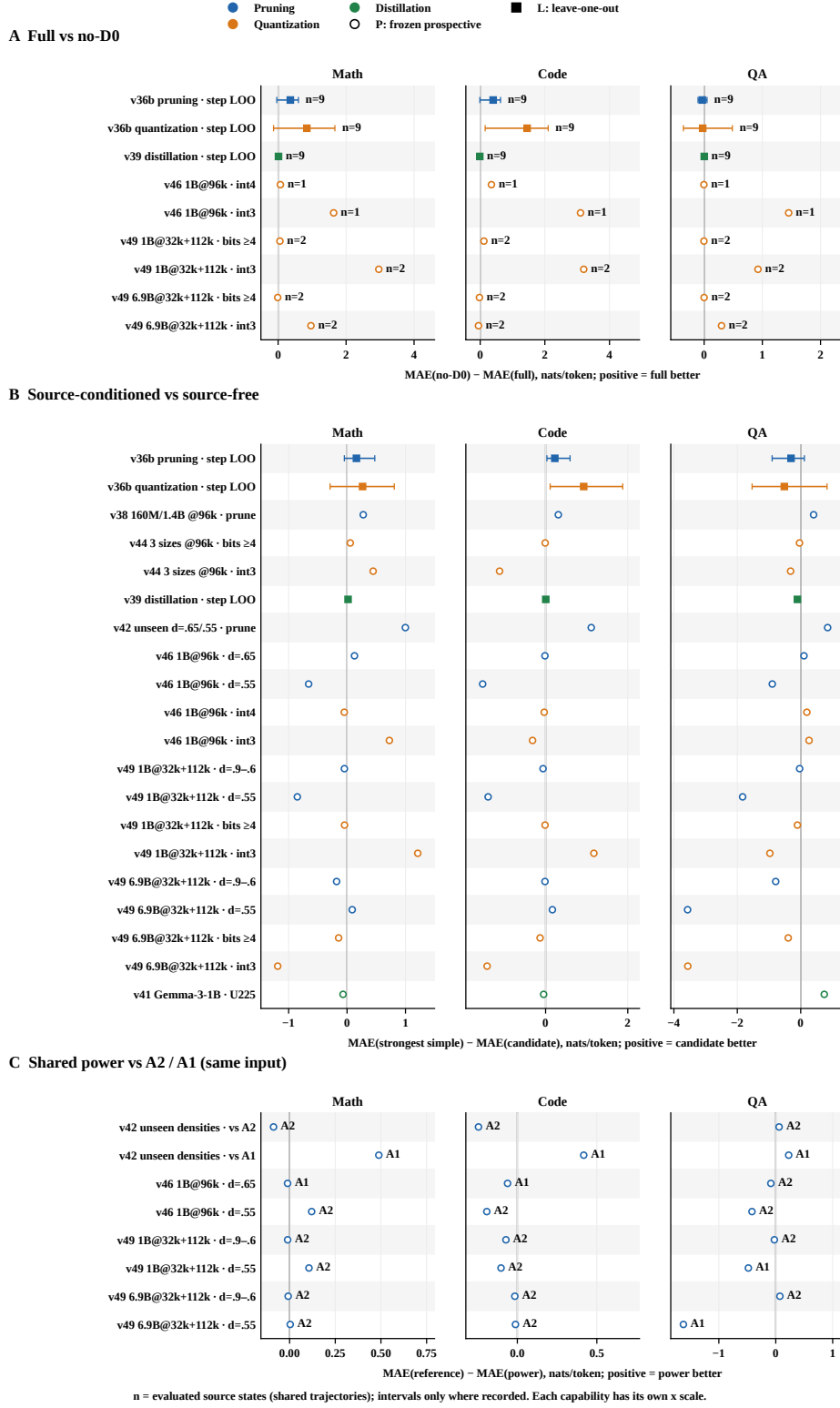


Figure 4: Paired improvements, zero line marked, positive means the first-named model is better. A: full model against its no- D_0 variant, per arm and capability; intervals only where a cluster bootstrap was computed (leave-one-stage-out panels), n = evaluated source states. B: source-conditioned candidate against the strongest source-free baseline for all twenty registered tests (hollow = frozen prospective, filled = leave-one-out). C: shared power form against the same-input alternatives A1 and A2 on the pruning configuration tests. Generated by plot_fig2_gains.py from v52 groups and the v36b/v39/v42/v46/v49 result files.

E Capability-Loss Measurement Checks

Capability specificity as a prediction test. On a six-model, five-density pruning panel (d from 0.9 to 0.7), the response on the primary benchmark of each capability was used to predict the response on the secondary benchmark of the same capability through a two-parameter affine map fit out of fold, and compared with the best cross-capability predictor chosen the same way. With whole models held out, the same-capability gain is +0.04 nats for math (95% model-bootstrap interval $[0.03, 0.05]$) and +0.14 for QA ($[0.02, 0.27]$); with whole densities held out it is +0.14 ($[0.03, 0.30]$) and +0.21 ($[0.02, 0.52]$). For code the cross-capability predictor is as good or better ($-0.06, [-0.16, -0.00]$ with densities held out), so code specificity is not established at this resolution (Fig. 5). Six model clusters give descriptive intervals only.

Same-capability agreement across the three operations. On Gemma-3-1B, six states fixed before the measurement (pruning at $d = 0.7$, RTN int4, and four distillation checkpoints of the uniform protocol at $U = 75$ and $U = 450$) were scored on the main-protocol probe and on an independent benchmark of the same capability (Table 21). Code agrees in sign and size on all six states. Pruning and quantization damage agrees in sign on all three capabilities, with sizes that differ on the short secondary probes (260 and 328 scored tokens for SVAMP and TriviaQA). The QA improvement under distillation on the primary probe (-0.7 to -1.5 nats) does not appear on TriviaQA ($+0.1$ to $+0.4$), so the distillation QA response reported in Section 3.4 is specific to the probe distribution, which shares its multi-hop format with the teacher pool, and should not be read as a general QA gain. The math response under distillation on SVAMP lies within that probe’s noise.

Table 21: Primary versus secondary benchmark on Gemma-3-1B (P3 check). Loss change from dense in nats per native token on the main-protocol probe / on an independent benchmark of the same capability (math: SVAMP; code: HumanEval; QA: TriviaQA; 128 samples, fixed probe seed, existing references). The six states were fixed before any secondary measurement. Scored completion tokens, primary / secondary: math 13198 / 260, code 4516 / 8393, QA 251 / 328.

State	Math	Code	QA
magnitude pruning $d = 0.7$	+1.31 / +0.51	+1.23 / +1.41	+0.83 / +3.30
RTN int4	+0.95 / +2.51	+0.60 / +0.75	+0.88 / +1.99
KD, $U = 75$, first milestone	+0.07 / -1.09	+0.06 / +0.06	-1.33 / +0.11
KD, $U = 75$, second milestone	+0.08 / +0.31	+0.12 / +0.07	-0.69 / +0.31
KD, $U = 450$, first milestone	+0.10 / -1.36	+0.06 / +0.06	-1.46 / +0.21
KD, $U = 450$, second milestone	+0.09 / -0.30	+0.11 / +0.07	-1.40 / +0.37

Loss and behavior. Loss-to-accuracy links were fit on GSM8K accuracy, MBPP pass@1, and TriviaQA exact match and token F1 across seven pruned models; all four outcomes are compatible with one cross-family sigmoid in the loss, with no resolved family-specific term. A lower QA loss does not by itself imply better QA behavior: the four pruned OLMo-3-32B cells whose QA loss lies below dense show no increase in exact match. This audit covers pruning only.

Reference conditions. For a subset of models the direct-answer and given-reasoning-answer conditional distributions were measured in addition to the full-solution reference; they differ from it for math and QA and coincide for code, where the reference is the function body, so the numbers in this paper are specific to the full-solution reference and are not evidence of format robustness.

The loss is the conditional cross-entropy of the reference completion given the prompt, summed over completion tokens and divided by the number of completion tokens, as in Eq. equation 1. Prompts and references are tokenized with the model’s own tokenizer and truncated to 1024 tokens; prompt tokens are masked out of the loss. All within-model responses ΔL_c are differences in native-token nats on identical token sequences. Because

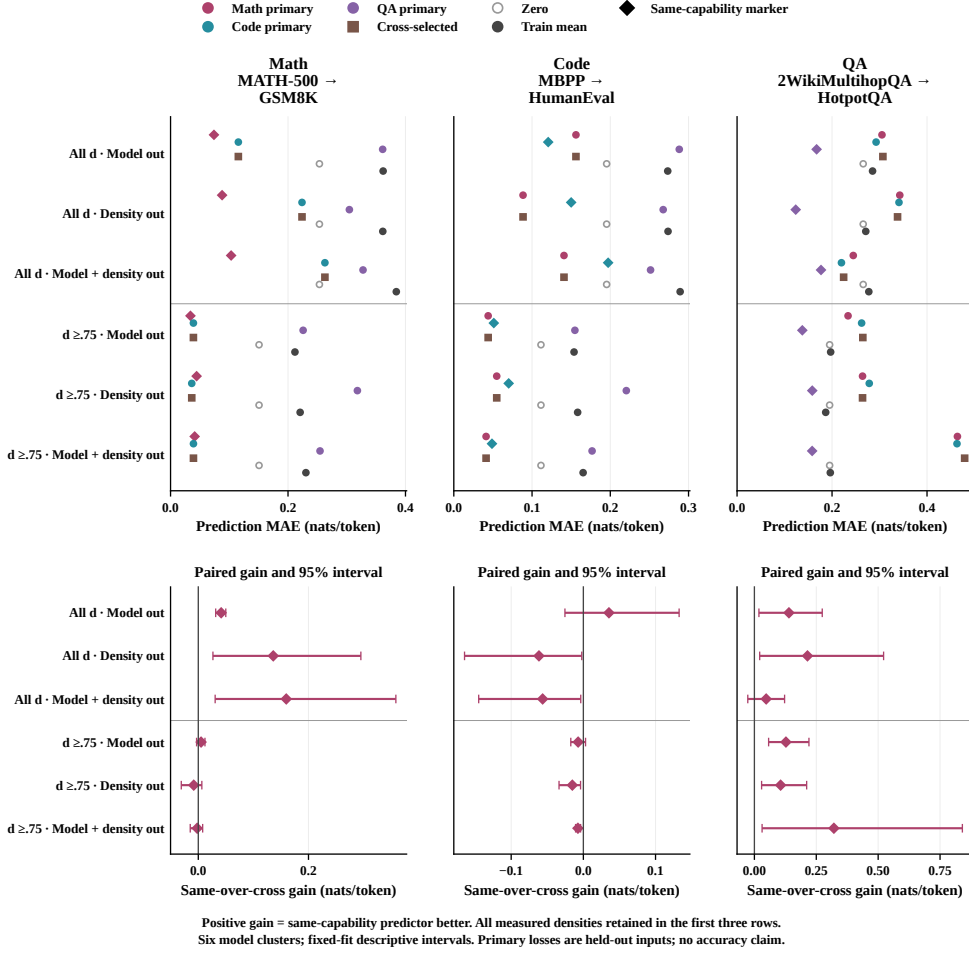


Figure 5: Measurement support for capability specificity. Top: out-of-fold MAE of predicting the secondary-benchmark response of each capability from the primary-benchmark response of the same capability (diamond) or of another capability, against zero and train-mean controls, on the six-model, five-density pruning panel; rows are hold-out protocols with all densities and with $d \geq 0.75$ only. Bottom: paired same-over-cross gain with 95% model-bootstrap intervals (six clusters, fixed out-of-fold fits). Generated by `plot_fig4_measurement.py` from `v26-loss-validity-pred/summary.json`.

native-token units differ across tokenizers, cross-family comparisons of absolute levels use per-byte normalization or are reported separately, and no cross-tokenizer quantity is treated as a common numerical scale. Negative responses are retained as measured; nothing is clipped at zero.

F Heterogeneous-Model Results

Pruning. Table 22 gives per-capability damage at two densities, the density at which math damage first exceeds one nat, and how QA ranks before that cliff. Magnitudes differ by two orders across series at the same density (OLMo-3-32B +0.01 against Gemma-3-27B +4.0 on math at $d = 0.7$). The ordering of capabilities is series-specific: in the Qwen3 sizes and in OLMo-3-32B, QA is the least-damaged capability at every pre-cliff density and improves, by up to -0.79 nats at Qwen3-4B; in every Gemma-3 size, in Gemma-4-31B, and in Muse-30B, QA is the most-damaged capability at the mildest densities (Gemma-4-31B: +2.89 QA

Model	ΔL_c at $d = 0.7$			math@0.5	d_{math}^*	QA least worst	min QA (d)
	math	code	QA				
Qwen3-0.6B	+0.49	+0.56	-0.26	+4.72	0.60	3/3 0/3	-0.26 (0.70)
Qwen3-1.7B	+0.24	+0.19	-0.51	+3.60	0.60	6/6 0/6	-0.90 (0.62)
Qwen3-4B	-0.00	-0.09	-0.79	+1.31	0.50	5/5 0/5	-1.96 (0.55)
Qwen3-8B [†]	+0.03	+0.10	-0.41	–	none by 0.60	4/4 0/4	-1.02 (0.60)
Qwen3-14B [†]	+0.06	+0.03	-0.45	–	none by 0.60	3/4 0/4	-0.80 (0.60)
Gemma-3-270M	+3.49	+4.04	+2.54	+11.21	0.70	0/2 1/2	+0.21 (0.90)
Gemma-3-1B	+1.31	+1.23	+0.83	+13.78	0.70	0/5 4/5	+0.19 (0.90)
Gemma-3-4B	+0.83	+1.47	+0.24	+7.00	0.60	1/3 2/3	+0.05 (0.90)
Gemma-3-12B	+0.68	+0.46	+0.54	+17.10	0.60	0/3 2/3	+0.07 (0.90)
Gemma-3-27B	+4.04	+4.69	+3.59	+15.21	0.70	1/2 1/2	+0.01 (0.80)
Gemma-4-31B	+0.68	+0.77	+2.89	+16.06	0.60	0/3 3/3	+0.22 (0.90)
Muse-30B	+0.17	+0.14	+0.20	+3.53	0.55	1/4 3/4	+0.02 (0.90)
OLMo-3-7B	+0.03	+0.03	+0.14	+0.22	0.35	4/8 3/8	-0.02 (0.55)
OLMo-3-32B	+0.01	+0.03	+0.02	+0.09	none by 0.30	7/10 0/10	-0.24 (0.30)

Table 22: Per-capability pruning damage on the heterogeneous panel (nats/token, native tokenizer units; within-model differences). d_{math}^* is the largest measured density with $\Delta L_{\text{math}} \geq 1$; models are grouped by series (Qwen3, Gemma-3, Gemma-4, Muse, OLMo-3); “math@0.5” is ΔL_{math} at $d = 0.5$; “QA least | worst” counts, over the pre-cliff densities, how often QA is the least-damaged and how often the most-damaged capability the last column is the most negative QA response and the density at which it occurs. [†]prospective additions measured at four densities only. Generated by v51_panel_tables.py from prune_losses.json.

Model	Series	ΔL_c at int4			ΔL_c at int3			bits
		math	code	QA	math	code	QA	
Qwen3-0.6B	Qwen3	+0.56	+0.73	+0.21	+8.83	+10.83	+6.61	8,6,5,4,3
Qwen3-1.7B	Qwen3	+0.40	+0.16	-0.21	+9.53	+9.46	+8.10	8,6,5,4,3
Qwen3-4B	Qwen3	+0.13	+0.19	+0.39	+10.73	+11.85	+5.99	8,6,5,4,3
Qwen3-8B [†]	Qwen3	+0.28	+0.48	-0.80	+12.86	+13.55	+9.45	8,6,5,4,3
Qwen3-14B [†]	Qwen3	+0.09	+0.31	-0.23	+9.30	+10.79	+6.45	8,6,4,3
Gemma-3-270M	Gemma-3	+1.99	+2.35	+0.88	+26.15	+27.94	+25.93	8,6,5,4,3
Gemma-3-1B	Gemma-3	+0.95	+0.59	+0.88	+15.33	+14.12	+9.80	8,6,5,4,3
Gemma-3-4B	Gemma-3	+0.72	+0.50	-0.27	+21.73	+19.24	+15.30	8,6,5,4,3
Gemma-3-12B	Gemma-3	+0.40	+0.32	-0.42	+20.19	+22.92	+20.75	8,6,5,4,3
Gemma-3-27B	Gemma-3	+0.33	+0.23	-0.67	+18.46	+17.15	+16.10	8,6,5,4,3
Gemma-4-31B	Gemma-4	+0.94	+0.58	+1.28	+20.35	+19.85	+14.90	8,6,5,4,3
Muse-30B	Muse	+1.58	+2.06	-0.11	+11.14	+11.18	+8.36	8,6,5,4,3
OLMo-3-7B	OLMo-3	+0.16	+0.21	-0.07	+4.70	+4.63	+2.76	8,6,5,4,3
OLMo-3-32B	OLMo-3	+0.04	+0.06	+0.02	+0.68	+1.22	+0.38	8,6,5,4,3

Table 23: Per-capability quantization damage (per-output-channel symmetric RTN, weights only; nats/token). int8 and int6 are within 0.01 nats of dense for every listed model and are omitted. [†]prospective additions. Generated by v51_panel_tables.py from quant_losses.json.

against +0.68 math at $d = 0.7$); OLMo-3-7B lies in between with differences of at most 0.15 nats. A noise screen sets the scale: the median $|\Delta L|$ at $d = 0.9$ is 0.013 nats, and 29 of the 62 nominal improvement cells across all arms exceed twice that median; the rest are treated as zero. Cliffs occur between $d = 0.5$ and 0.7 for Qwen3 and Gemma, at 0.55 for Muse-30B and 0.35 for OLMo-3-7B, and had not occurred by $d = 0.3$ for OLMo-3-32B. Robustness is not monotone in size within a series (Gemma-3-12B is the most robust Gemma-3 size; 27B collapses earlier). We read this as heterogeneity a predictor has to reproduce, not as an invariant.

Quantization. Table 23 covers 14 models. Int8 and int6 stay within 0.01 nats of dense. Int4 damages small models (Gemma-3-270M +2.0, +2.4, +0.9) and barely touches large

ones (Gemma-3-27B +0.33, +0.23, -0.67); a QA improvement under int4 recurs in five models and clears the noise screen. Int3 collapses 12 of the 14 models by 4.6 to 28 nats; OLMo-3-7B loses 2.8 to 4.7 and OLMo-3-32B only 0.4 to 1.2, so the collapse is common but not universal, and OLMo-3 is the robust series under both interventions. Within the tested bit-widths a learned exponential $q_c e^{-kb}$ fits held-out bits better than the fixed $q_c 4^{-b}$ shape (pooled MAE 6.7 against 7.6 nats), which says the fixed base is wrong for this quantizer and nothing more; the squared step-size ratio between 4 and 5 bits is $(7/15)^2 \approx 0.22$ (Appendix A), and a loss ratio would follow only under error-distribution assumptions we do not test.

Distillation. Black-box traces from two API teachers (600 per domain) train Gemma-3 students under a full-trace and an answer-only recipe in a paired design. The answer-only recipe hurts math more (+0.19 to +0.22 against +0.09 to +0.12 nats relative to the initial student) while QA improves under every recipe (-0.58 to -1.44); because the recipes differ in supervised tokens and update counts, this pairing does not isolate reasoning coverage, and the controlled comparisons are those of §3.4.

What the endpoint does and does not say. The same heterogeneity appears on independent secondary benchmarks for math and QA, where the primary-benchmark response predicts the secondary one better than any other capability’s response does, with intervals excluding zero in most hold-out protocols; for code the cross-capability predictor is as good or better (Appendix E, Fig. 5). A lower QA loss does not by itself mean better QA behavior: in the four pruned OLMo-3-32B cells whose QA loss lies below dense, exact-match accuracy does not rise. “QA improves” in this paper refers to the loss.

F.1 Scale and family structure

Within-family robustness is non-monotonic in two families independently. Gemma-3 robustness peaks at 12B (27B collapses far earlier, +4.0 vs. +0.68 at $d = 0.7$), mirroring the prior ability-space observation of a Qwen3 14B peak. A mid-scale robustness peak is a reproducible pattern of scaling-down.

A candidate predictor, honestly retired. The top-concentration statistic κ_c tracked robustness within the Qwen3 ladder and was pre-registered as a predictor; the full grid retires it (cross-family inversion: OLMo, the most robust family, has the lowest concentration; within-Gemma top-end violation).

Matched-size panel (~ 30 B). At matched scale the family coefficient dominates: ΔL_{math} at $d = 0.7$ spans +0.02 (OLMo-3-32B), +0.17 (Muse-30B), +0.68 (Gemma4-31B), +4.04 (Gemma3-27B). OLMo-3-32B survives 70% pruning with $\leq +0.63$ nats on each capability. Family identity is a first-order determinant of prunability on this panel; which training-data or recipe properties carry it was not measured.

G Mechanistic Analyses

G.1 Capabilities occupy distinct regions of gradient space

Convergent validity (block structure). For each of nine benchmarks we measure a gradient signature (the benchmark-conditional diagonal Fisher) and compare all pairs. After removing the component shared across benchmarks, signatures of same-capability benchmarks are positively aligned (cosine +0.36 to +0.45) while signatures of different capabilities are anti-aligned (-0.23 to -0.25), in each of the ten models from five series for which the block analysis was run (Figs. 6–15); the C4 control sits between blocks. (preliminary evidence: point estimates; example-bootstrap, permutation, and prompt-stability tests are left to future work) The mathematics benchmarks cluster regardless of format, and within-block fine structure matches intuition (SVAMP, grade-school word problems, is the loosest member of the math block). A projected-gradient SVD version gives the same verdict in subspace terms: principal-angle similarity within capability exceeds cross-capability similarity by 2–4 $\times$.

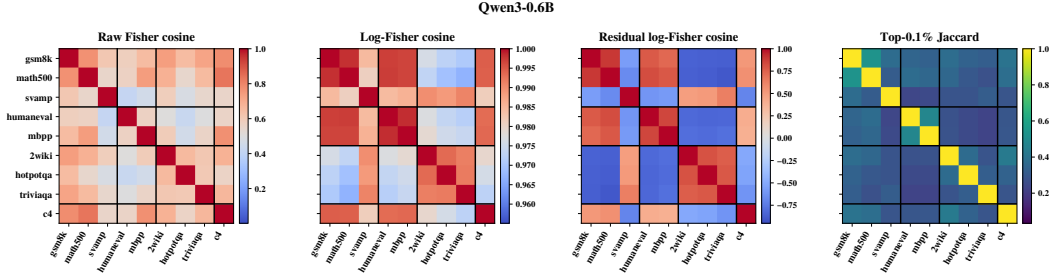


Figure 6: Capability regions in gradient space for Qwen3-0.6B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

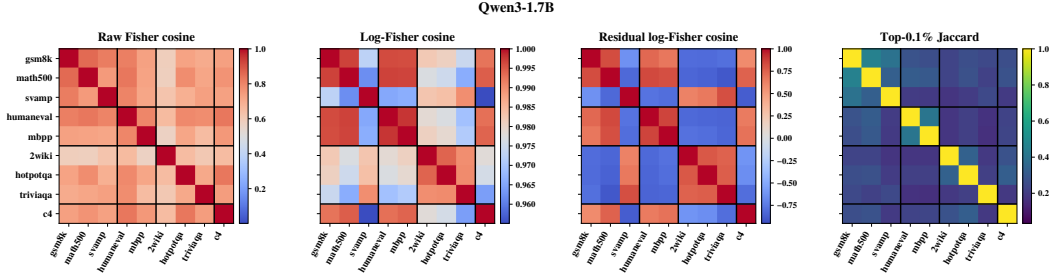


Figure 7: Capability regions in gradient space for Qwen3-1.7B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

Selective ablation (preliminary evidence: single model; random, magnitude-matched, and global-Fisher controls are left to future work). A first targeted test: select the top 0.05% of weight coordinates by each capability’s residual Fisher signature (excluding coordinates shared with other capabilities) and zero them. The damage matrix is nearly diagonal (Gemma-3-1B; rows ablate, columns measure, nats/token):

Ablated region	math	code	QA
math	+4.63	+0.43	+0.05
code	+0.42	+7.14	+0.00
QA	+0.01	+0.00	+2.29

Zeroing 0.05% of weights, selected from gradients alone, selectively removes one capability while leaving the others nearly untouched (the QA ablation moves code by +0.0001). Pending multi-model replication and the control ablations above, we describe the regions as causally implicated and partially separable; we do not call them proven causal carriers; if confirmed, ablation itself is a form of targeted scaling-down.

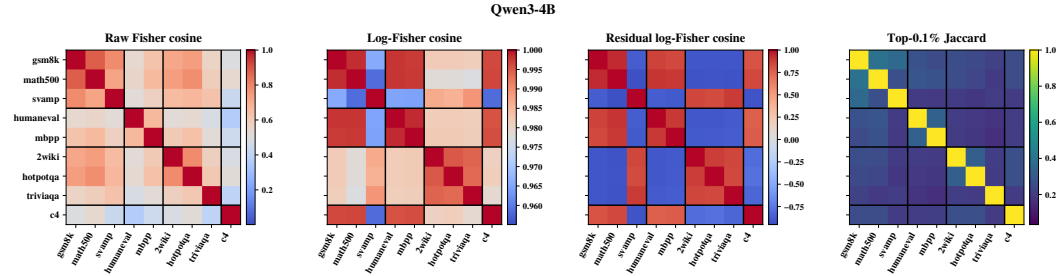


Figure 8: Capability regions in gradient space for Qwen3-4B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

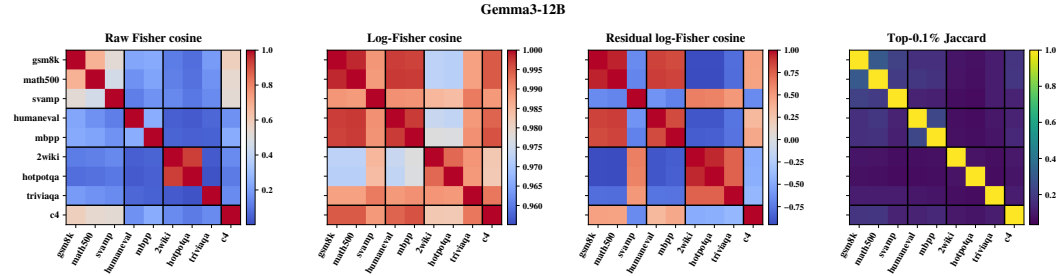


Figure 9: Capability regions in gradient space for Gemma3-12B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

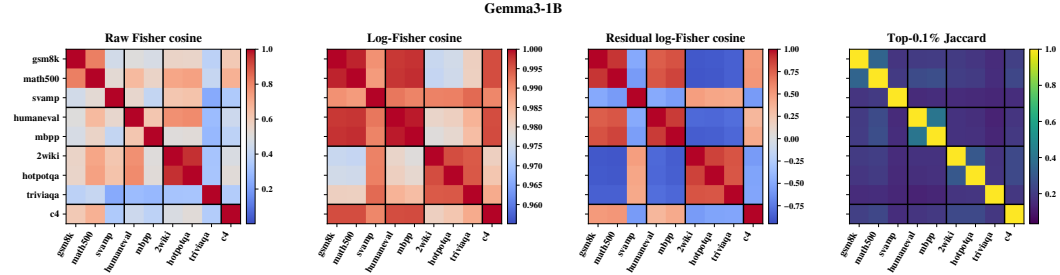


Figure 10: Capability regions in gradient space for Gemma3-1B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

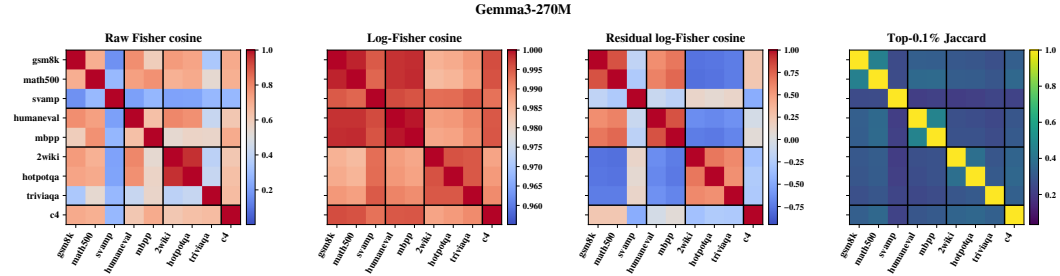


Figure 11: Capability regions in gradient space for Gemma3-270M: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

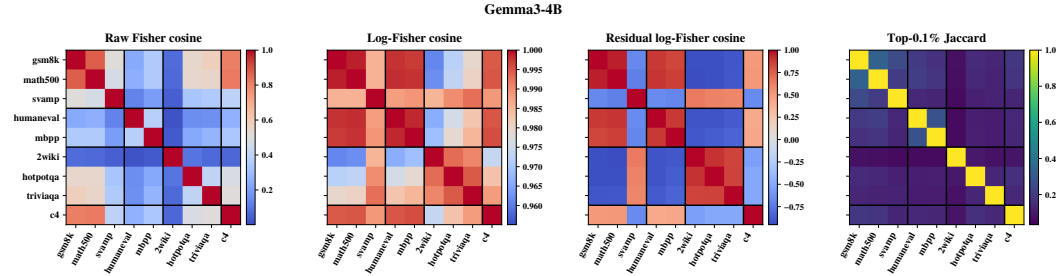


Figure 12: Capability regions in gradient space for Gemma3-4B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

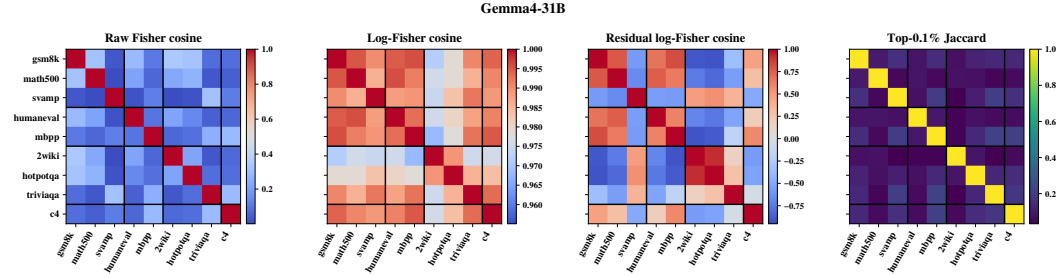


Figure 13: Capability regions in gradient space for Gemma4-31B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

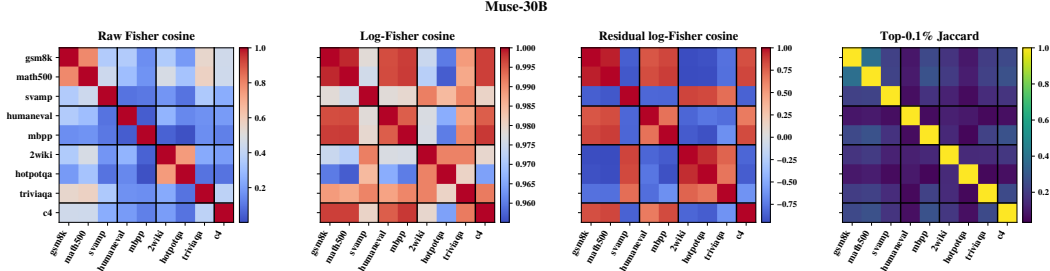


Figure 14: Capability regions in gradient space for Muse-30B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

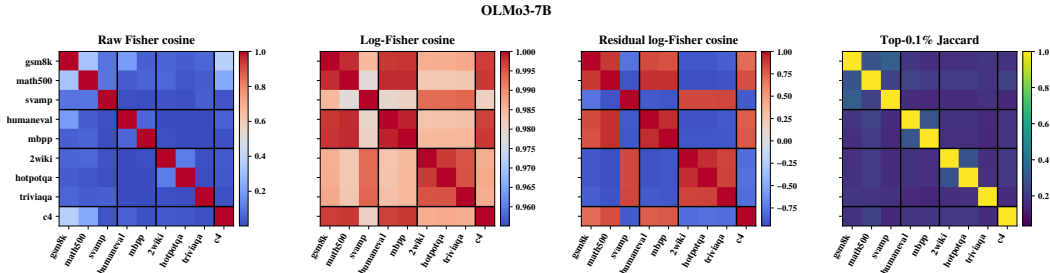


Figure 15: Capability regions in gradient space for OLMo3-7B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

G.2 The pruning law

The mechanism form below fits the entire pre-cliff grid. With the measured alignment term subtracted, the curvature residual follows a power law in the deleted capability mass $m_c(d)$:

$$\Delta L_c(d) = \underbrace{\bar{g}_c^\top \delta w_d}_{\text{measured}} + B \left[\underbrace{m_c(d)}_{\text{measured}} \right]^\gamma, \quad (7)$$

with in-sample $R^2 = 0.82\text{--}0.996$ over 24 capability $\times$ model fits. The held-out battery (1000-resample cell bootstrap) is where the form is tested: in leave-one-family-out transfer the shared- γ mechanism form attains MAE 2.17 nats versus 10.3 for per-capability raw sparsity and 8.2 for the direct-damage baseline (both CIs exclude zero), so the deleted-mass coordinate is what transfers across families; it also beats the direct-damage baseline in fit-shallow/predict-deep (3.11 vs. 4.24) and raw sparsity in leave-largest-model-out (1.41 vs. 2.15), while its edge over raw sparsity within-model does not reach significance (without the first-order term, QA is unfittable, $R^2 \leq 0.67$; with it, QA joins the law). Two structural results: (i) in the deleted-mass coordinate the exponent concentrates at $\gamma \in [1.2, 2.2]$ across all families, versus 3.2–8.7 in the raw $(1-d)$ coordinate, which is what makes the form portable; and (ii) within a model, γ is shared across capabilities (e.g. Qwen3-0.6B: 1.41/1.56/1.43; Gemma3-4B: 1.66/1.71): capability differences appear to be carried by the two measured quantities, and the fitted structure is two parameters per model. This mechanism form uses gradient information of the dense model and is therefore not a K0 predictor; it is reported as an upper comparator for the reduced forms of the main text, not as a bound.

G.3 The signed first-order term predicts, in both directions

The alignment term $\bar{g}_c^\top \delta w$ is the family coefficient that carries sign. Its sign predicts the presence or absence of the capability-selective improvement in each case measured: the QA term of Qwen3 is negative and QA improves under pruning (predicted -0.83 vs. measured -0.79 at 4B, $d = 0.7$); the QA term of Gemma is positive and no improvement exists at any size; the term for OLMo-3-7B is positive (mild degradation, prediction $+0.015$ = measurement at $d = 0.8$, code) while the term for OLMo-3-32B is negative, and deep-pruning QA improvement duly appears. At the level of individual model $\times$ capability $\times$ density cells in the mild regime, sign prediction accuracy is 67.6% (Wilson 95% CI $[58.3, 75.7]$, $n=108$), above chance, with the family-level pattern (Qwen and OLMo-32B QA improvements, Gemma none, an OLMo size flip) called in advance. Beyond the mild regime, first- plus second-order predictions fail by orders of magnitude at the cliff in each family: the collapse is non-perturbative in loss space, confirming the two-regime structure of equation 7.

G.4 The cliff is a geometric threshold crossing

Measuring the capability geometry of the pruned models accounts for the location of the cliff on the three models tested. The capability-region mass retention (the fraction of the capability-exclusive top coordinates of the dense model still prominent after pruning) declines smoothly from the first densities (no cliff in geometry) while loss is still flat; the behavioral cliff occurs where retention crosses a critical value. The critical value is family-specific (Gemma: 0.035–0.10 across 1B/4B and all capabilities; Qwen: 0.13–0.23) but consistent within a family across sizes and capabilities: the cliff boundary $d_c^*(N_0, f)$ reduces to one critical-retention constant per family, turning the least elegant fitted parameter of the two-regime law into a measurable, smoothly evolving order parameter. (preliminary evidence: three models, two families; a multi-model extension is left to future work)

H Recovery and Cross-Method Analyses

H.1 Recovery arm: redeemable damage and the fitted plateau

Recovery training on general text fits the recovery law cleanly (Gemma3-270M pruned to $d = 0.6$; $R^2 \approx 1.0$): $\beta'_{\text{math}} = 1.23$, $\beta'_{\text{code}} = 1.02$, with fitted plateaus under the fixed LoRA recipe and the largest measured budget of 16M tokens, $L_c^{\text{plateau}} - L_c^{\text{dense}}$, of 1.21 (math) and

1.93 (code) nats; about 80% of this pruning damage is redeemed within 8M tokens and the remainder persists up to the largest budget measured under this recipe, while QA recovers completely. The plateau is a fitted quantity at 16M tokens, not an established asymptote, and the saturating form does not describe the non-monotone aligned-data trajectory below. (preliminary evidence: one model, one density, one recipe; with three-point ladders the largest-budget extrapolation of the saturating form beats a carry-forward baseline in only 4 of 9 fits, so the ladder is too sparse to claim extrapolation, and a denser multi-model grid is left to future work) Two mechanism results: recovery on aligned data (teacher traces) heals substantially faster at small budgets (math 2.76 vs. 3.47 at 0.5M tokens), as the gradient-overlap prediction requires, but over-training on narrow aligned data re-damages the model (QA 4.95 $\rightarrow$ 9.75 from 0.5M to 8M tokens), while general-text recovery is monotone. Aligned recovery is fast at small budgets and harmful when over-trained on narrow data; general recovery is slow and monotone on this model. Quantization damage past the bit cliff is markedly less redeemable under the same recipe (residuals $\approx$ 2.6–2.8 after int3) than pruning damage at matched severity.

H.2 Resource-ratio models and method selection

Table 24 gives the per-policy results behind Fig. 16: mean regret against the measured oracle, agreement with the oracle’s method, the share of no-clear-winner cells, and the share of cells in which a fixed-method policy cannot meet the budget (it then falls back to the dense source, which is why its regret can be negative). Generated by v60_selection_maps.py.

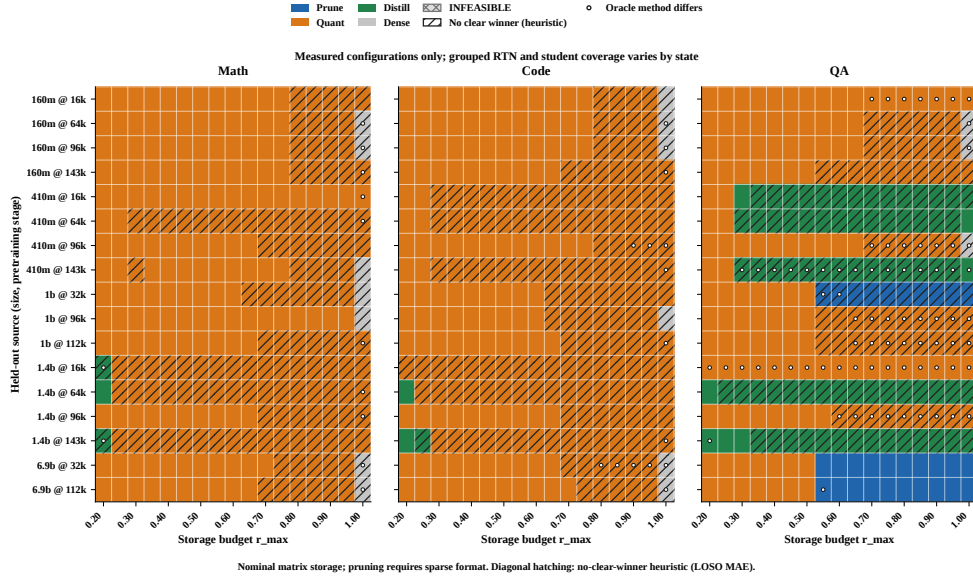


Figure 16: Capability-specific selection maps on held-out Pythia states. For each held-out state (rows, ordered by size and stage) and nominal storage budget $r_{\max}$ (columns), the colour is the method whose law-predicted loss is lowest among the measured configurations within budget (pruning $r = d$, per-channel RTN $r = b/16$, grouped RTN $r = (b + 16/g)/16$, distillation to a smaller same-stage student $r = N_S/N_0$; the dense source at $r = 1$); every law is refit with the state held out. Grey cross-hatching marks cells with no feasible configuration for a policy; diagonal hatching marks cells in which the predicted gap between the two best methods is smaller than the larger of their leave-one-state-out errors (a heuristic); a marker shows where the measured oracle chooses a different method. Nominal pruning storage requires a sparse format. Generated by v64_selection_feasible.py.

Table 24: Feasible leave-one-state-out selection. Coverage and oracle-method agreement are percentages; regret is in nats. Own/common refer to the subsets defined below.

Policy	Objective	Coverage	Regret (nats)		Agreement (%)	
		(%)	Own	Common	Own	Common
MAP	Math	100.0	0.1936	0.2018	95.8	94.5
MAP	Code	100.0	0.1743	0.1112	94.8	96.4
MAP	QA	100.0	0.4184	0.1578	73.4	67.3
MAP	All (max)	100.0	0.0288	0.0174	93.8	94.5
ORACLE	Math	100.0	0.0000	0.0000	100.0	100.0
ORACLE	Code	100.0	0.0000	0.0000	100.0	100.0
ORACLE	QA	100.0	0.0000	0.0000	100.0	100.0
ORACLE	All (max)	100.0	0.0000	0.0000	100.0	100.0
prune-only	Math	55.7	0.5875	0.2896	0.0	0.0
prune-only	Code	55.7	0.6218	0.3098	5.0	0.0
prune-only	QA	55.7	0.7245	0.4469	44.1	9.1
prune-only	All (max)	55.7	0.6731	0.3295	1.2	0.0
quant-only	Math	100.0	0.2089	0.2018	96.5	92.7
quant-only	Code	100.0	0.1943	0.1112	94.1	96.4
quant-only	QA	100.0	0.4905	0.5735	47.8	7.3
quant-only	All (max)	100.0	0.0428	0.0178	93.1	90.9
distill-only	Math	33.2	0.2968	0.3225	1.0	0.0
distill-only	Code	33.2	0.3366	0.3626	3.1	0.0
distill-only	QA	33.2	0.0564	0.0665	83.3	83.6
distill-only	All (max)	33.2	0.3332	0.3606	3.1	0.0
CHEAPEST	Math	100.0	4.5839	2.0931	79.9	43.6
CHEAPEST	Code	100.0	5.1745	2.2502	78.9	47.3
CHEAPEST	QA	100.0	4.3334	1.8392	64.7	58.2
CHEAPEST	All (max)	100.0	5.1782	2.2700	78.2	43.6

Coverage is feasible cells / 289. Own averages only that policy’s feasible cells; common averages the same 55/289 cells feasible for all six policies, including CHEAPEST. Agreement uses those same respective subsets. Empty subsets are N/A. An empty candidate set returns INFEASIBLE with no source fallback. Dense ($r = 1$) is eligible for MAP/ORACLE/CHEAPEST only at $r_{\max} = 1$. Fixed policies retain only their named method; quant-only pools channel and grouped RTN. Each deployed endpoint is an absolute loss: source dense plus response for pruning/RTN; student dense plus student post-training change for KD. The multi objective is $\max_c(L_c - L_{0c})$. The unchanged no-clear-winner heuristic uses pooled LOSO MAEs and is not calibrated uncertainty. Five-bit folds remain underidentified; measurement coverage and dense anchors vary by arm.

Table 25: Measured candidate coverage by source state. Each method column gives configuration count / minimum available storage ratio r .

Source state	Prune	Channel RTN	Grouped RTN	KD students	Min r : quant
160m @ 16k	4 / 0.6000000	4 / 0.1875000	9 / 0.1914062	0 / N/A	0.1875000
160m @ 64k	4 / 0.6000000	4 / 0.1875000	0 / N/A	0 / N/A	0.1875000
160m @ 96k	4 / 0.6000000	4 / 0.1875000	0 / N/A	0 / N/A	0.1875000
160m @ 143k	6 / 0.5500000	4 / 0.1875000	9 / 0.1914062	0 / N/A	0.1875000
410m @ 16k	4 / 0.6000000	4 / 0.1875000	9 / 0.1914062	1 / 0.2812500	0.1875000
410m @ 64k	4 / 0.6000000	4 / 0.1875000	0 / N/A	1 / 0.2812500	0.1875000
410m @ 96k	6 / 0.5500000	4 / 0.1875000	0 / N/A	0 / N/A	0.1875000
410m @ 143k	4 / 0.6000000	4 / 0.1875000	9 / 0.1914062	1 / 0.2812500	0.1875000
1b @ 32k	7 / 0.5500000	5 / 0.1875000	0 / N/A	0 / N/A	0.1875000
1b @ 96k	2 / 0.5500000	2 / 0.1875000	3 / 0.1953125	0 / N/A	0.1875000
1b @ 112k	7 / 0.5500000	5 / 0.1875000	0 / N/A	0 / N/A	0.1875000
1.4b @ 16k	4 / 0.6000000	4 / 0.1875000	9 / 0.1914062	2 / 0.0703125	0.1875000
1.4b @ 64k	6 / 0.5500000	4 / 0.1875000	0 / N/A	2 / 0.0703125	0.1875000
1.4b @ 96k	4 / 0.6000000	4 / 0.1875000	0 / N/A	0 / N/A	0.1875000
1.4b @ 143k	4 / 0.6000000	4 / 0.1875000	9 / 0.1914062	2 / 0.0703125	0.1875000
6.9b @ 32k	7 / 0.5500000	5 / 0.1875000	0 / N/A	0 / N/A	0.1875000
6.9b @ 112k	7 / 0.5500000	5 / 0.1875000	0 / N/A	0 / N/A	0.1875000
Total	84	70	57	9	—

N/A means no measured candidate. Each state also has one dense source, minimum $r = 1$; it is excluded from method counts. KD has 0, 1 or 2 smaller same-stage measured students. Student reuse across sources counts as separate alternatives, not independent measurements. Quant pools channel and grouped RTN. Ratios: pruning $r = d$, channel RTN $r = b/16$, grouped RTN $r = (b + 16/g)/16$, KD $r = N_S/N_0$. Storage is nominal transformer-matrix storage; pruning requires a sparse format (index overhead excluded). No missing configurations are imputed. Coverage is measured availability, independent of the no-clear-winner heuristic.

Resource-ratio models. A method-agnostic response in the storage or active-parameter ratio was tested on the heterogeneous panel: at the common storage ratio 0.5, pruning to $d = 0.5$ costs median +5.9, +8.2, and +2.9 nats while int8 quantization costs none; over 75 storage-matched pairs (14 with both sides pre-cliff) the mean absolute method gap is 9.3 nats with sign agreement 0.61, and the two shared-law hypotheses fit on pre-cliff cells reach held-out sign accuracies of 0.51 and 0.37. These models are rejected; whether another single form could unify the arms is untested. A capability- and cost-aware method selection would use predictors like these as loss constraints; the offline replay with frozen predictions below quantifies the decision cost of prediction error on the heterogeneous panel, and the held-out selection validation on the Pythia panel is in Section 5 and Appendix H.2.

H.3 Re-analysis of an earlier ability-space grid

A prior internal study fit compression laws directly on Rasch abilities from ~ 100 -item panels over a 774-cell Qwen3 grid and reported that no derived law had held-out predictive power. Our re-analysis attributes much of this to the measurement layer: (1) the $r=1$ falsification anchor was driven almost entirely by the answer-only recipe (stratified, the anchor holds and held-out error drops from $9.0\times$ to $1.8\times$ the noise floor); (2) same-benchmark evaluation inflated QA gains by roughly one third; (3) discrimination heterogeneity voids interval-scale ability while rankings survive ($\rho \geq 0.98$); (4) item-level damage is smooth pre-cliff, ruling out discrete skill deletion at the item level. These findings fixed the protocol of §2.1.

I Reproducibility and Prediction Registration

Data cut-off. All numbers in this version come from measurements completed by 19:47 EDT on 10 September 2026 (repository commit 3b5feae); frozen predictions are cited with the commit that recorded them, and nothing measured after that time is used.

I.1 Registration and provenance codes

Each prediction test in Table 1 follows one order. The functional forms, the development set, and each standardization and coefficient are fixed and committed first. For a new source the dense capability loss is then measured, since it is a permitted pre-compression input; the predictions for each configuration and candidate are written to a versioned file and committed to the repository, with the commit hash recorded; and only then is the compressed or distilled outcome measured. Information budgets are reported separately: K0 uses pre-compression information only, K1 adds one pre-specified target-side calibration point, and post-hoc rescaling on target outcomes is reported only as an oracle diagnostic. Each comparison carries the four-part origin code defined with Table 1 and Appendix D; the label of a frozen prediction never depends on its outcome. Register files (paths in the repository; content hashes in docs/MANUSCRIPT_REWRITE_MAP.md): v36b-input-comparison/summary.json (leave-one-step-out source axis, both arms), v38-prospective/register.json (frozen step-96k coefficients), v40-prune-strength/register.json (shared power form and source-free curves), v41-distill-newpool/summary.json (the four frozen pool forms), v46-p1-newsource/predictions_frozen.json (1B@96k), v49-plv2/register.json and predictions_*.json (1B and 6.9B pairs, protocols A and B). When a diagnostic overturns an earlier headline, the earlier claim is downgraded in the text and the prediction itself stays as committed.

I.2 Information budgets

Budget	Information available	Cost to disclose
K0, metadata only	N_0 , public D_0 , architecture, intervention configuration	metadata availability; D_0 may be undisclosed
K0 + dense anchor	the above plus the dense capability losses	probe count, one forward pass per probe, native loss units
K1	L_0 , measured once before compression the above plus one pre-specified measurement of the target at one compression configuration	the calibration configuration and its compression cost
Oracle diagnostic	post-hoc rescaling on the target outcomes	none; diagnoses form misfit, never a prediction

Table 26: Information budgets. Unknown D_0 stays unknown; handling of missing inputs is fixed on the development panel and never inferred from the target response. Predicting L_0 from (N_0, D_0) is the task of dense scaling laws and is not attempted here.

I.3 Reproduction

All tables are generated from machine-readable result files by CPU-only scripts that run in seconds and require neither model weights nor a GPU; the repository mirrors the scripts and the result files they read, and the main prediction table is produced by a single command. Model measurements are reproducible from the recorded revisions, probe seeds, and protocols above.